6. Impacts of climate change on Oregon's coasts and estuaries

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Summary and Knowledge Gaps

Earth's changing climate is expected to have significant physical impacts along the coast and estuarine shorelands of Oregon, ranging from increased erosion and inundation of low lying areas, to wetland loss and increased estuarine salinity. The environmental changes associated with climate change include rising sea levels, increased occurrences of severe storms, rising air and water temperatures, and ocean acidification. The combination of these processes and their climate controls are important to beach and property erosion, flood probabilities, and estuarine water quality, with the expectation of significant changes projected for the 21st century.

Coastal change and flood hazards along the Oregon coast are caused by a number of ocean processes, each of which has significant climate controls such that the severity and frequency of the hazards in the future can be expected to increase. There is near certainty that the rate of sea-level rise will increase in the future as a result of global warming, with the potential of greater than a 1.0 meter increase in sea level by 2100. Evaluating the consequences of intensified and more frequent hazards is complicated by Oregon's tectonic setting, with there being significantly different rates of land uplift along the coast. Taken together, the variable rate of uplift plus the present-day rate of sea level rise, some stretches of the coast are submerging as the sea level rise is greater than the tectonic uplift, whereas other areas are emerging where the reverse is true. The prospects are that with accelerated rates of sea level rise, the entire coast will eventually be submerging and experience significantly greater erosion and flood impacts than at present day.

Another long-term trend is increasing storm intensities and the heights of the waves they generate. In addition, the periodic occurrence of major El Niños in the future will compound the impacts of increasing sea levels and waves, resulting in severe episodes of coastal erosion and flooding, as experienced during the El Niño winters of 1982-83 and 1997-98. At present it is not known whether or not El Niño intensity and frequency will increase under a changing climate. With these multiple processes and their climate controls having important roles in causing erosion and flooding along the Oregon coast, it is challenging to collectively analyze them with the goal of providing meaningful assessments of future coastal hazards during the next several decades.

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Coastal infrastructure will come under increased risk to damage and inundation under a changing climate with impacted sectors including transportation and navigation, coastal engineering structures (seawalls, riprap, jetties etc.) and flood control and prevention structures, water supply and waste/storm water systems, and recreation, travel and hospitality.

It is likely that regional coastal climate change will result in changes in the intensity and timing of coastal upwelling, shifts in temperatures and dissolved oxygen concentrations, and alteration of the carbonate chemistry (ocean acidification) of nearshore waters. The combination of these meteorological and nearshore ocean changes will exert stress on the communities of near-coastal and estuarine organisms. The range of community responses to the climate change stressors may include elevational shifts in the distribution of submerged aquatic vegetation, disruption of shell formation for calcifying organisms, alteration of the phenology of phytoplankton blooms, shoreward migration of tidal marshes, and increased colonization by non-indigenous aquatic species.

Unfortunately, significant knowledge gaps remain, impairing our ability to accurately assess the impacts of climate change along our coast and estuarine shorelands. For example, the uncertainty of future global sea level rise is significant with credible projections ranging from less than 0.5 m to as much as 2.0 m by 2100. At present we do not conclusively understand the climate controls on increasing storm intensities and wave heights and therefore have a very limited ability to project future trends in coastal storm impacts. The magnitude and frequency of major El Niños has significant implications for the state of Oregon; however, at this time we are unable to assess whether or not these will increase in the future due to climate change. Further, the long-term time-series data necessary to definitively identify perturbations of estuarine communities that can be attributed to anthropogenic climate change are lacking and therefore our understanding of anticipated shifts remain largely speculative.

6.1 Introduction

The Oregon coast is approximately 500 kilometers long, extending from the mouth of the Columbia River in the north to the Klamath Mountains and the California border to the south. The rugged character of this picturesque coast results directly from its geologic setting, involving the collision of Earth's tectonic plates off its shore, with the subduction of the oceanic plates beneath the continent (Figure 6.1). There are 17 large rocky headlands (Figure 6.2), with the coast otherwise consisting primarily of a series of littoral cells, stretches of beach that are confined between bounding promontories. Each stretch of shore is in effect a pocket beach, although they range widely in their along-coast lengths, geometry, and varying capacity of their beaches to act as buffers between storm waves and the backing sand dunes or sea cliffs along which infrastructure is located. In addition there are 43 estuaries and tidal creeks along the Oregon coast (11 of which have jettied entrances), but the majority of them are relatively small (< 10 km^2; Lee and Brown, 2009). The geomorphology of the estuaries is diverse, including river-dominated drowned river mouths, tidal dominated drowned river-mouths, bar-built coastal lagoons, and numerous tidal creeks.

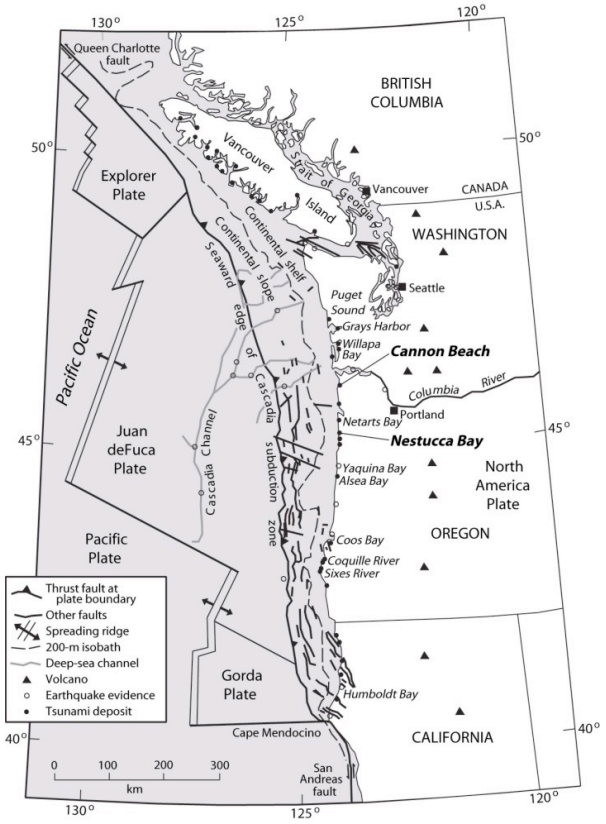


Figure 6.1 The tectonic setting of the Pacific Northwest, with the collision and subduction of the ocean plates beneath the continental North American plate (Satake et al., 2003).

While the tectonic setting of the Oregon coast has fundamentally determined the geological framework of its headlands, beaches and estuaries, it is the dynamics of ocean and estuarine

processes that determine the evolution of this coast. Many of these processes are controlled by climate and Earth's changing climate is expected to have significant physical impacts along the coast and estuarine shorelands of Oregon, ranging from increased erosion and inundation of low lying areas, to wetland loss and increased estuarine salinity. The environmental changes associated with climate change include rising sea levels, increased occurrences of severe storms, rising air and water temperatures, and ocean acidification. The combination of these processes and their climate controls are important to beach and property erosion, flood probabilities, and estuarine water quality, with the expectation that these impacts along the Oregon coast will significantly increase throughout the 21st century.

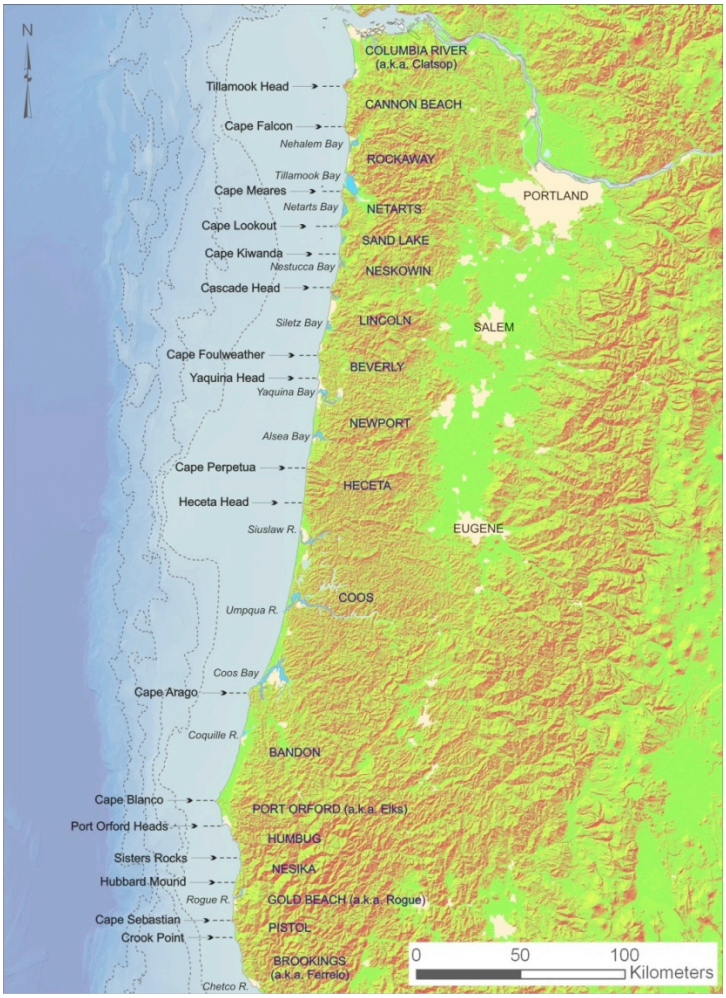


Figure 6.2. The 17 major headlands (bold font with arrows), major estuaries (light font), and littoral cell names (capital letters) of the Oregon coast.

Due to the significance of existing coastal problems and the potential for future increases in these problems, there has been increased focus by coastal scientists and engineers directed toward documenting the impacts of climate change along our coasts. In this chapter we summarize our present understanding of the climate controls on coastal and estuarine processes that are important to Oregon. While this review represents a progress report in that much of this

research is still underway, these investigations have advanced to the degree that we can document the basic changes in important environmental variables (e.g., sea levels, wave heights, shoreline position), supporting preliminary assessments of the expected magnitudes of future problems along the Oregon coast and the extent to which they potentially could increase due to Earth's changing climate.

In the following sections we attempt to summarize the most recent literature documenting historical changes as well as what may be expected to occur in response to climate change. Where little information is available we draw preliminary conclusions about the potential for specific impacts. When possible we highlight what research is needed to bridge knowledge gaps to improve our ability to identify climate change impacts more precisely, ultimately allowing for future projections.

6.2 Sea Level Rise Trends

Recent assessments of climate change have documented that globally the average level of the sea rose throughout the 20th century, with the total increase having been on the order of 15 to 20 centimeters, with an average sea level rise (SLR) rate of about 1.5 to 2.0 mm/year. More precisely, a recent analysis by Holgate (2007) yielded a global average rate of 1.74 mm/year (± 0.16 mm/year) during the 20th century. These assessments were based on records derived from tide gauges around the world, which measure the hour-to-hour changes in the water levels caused by tides, but when averaged for the entire year yield a value for the mean sea level for that year. Over the decades such analyses for each year document any progressive changes in the sea level.

A number of U.S. tide gauges have been in operation for more than a century such as those in San Francisco Bay and Seattle, while several in Europe provide records of changing sea levels that date back to about 1700 (Woodworth, 1999). Analyses based on those very long records demonstrate acceleration in SLR during the 19th century (e.g., Gornitz and Solow, 1991). This increase in the rate of SLR since the late 19th century is consistent with geological records (e.g., Donnely et al., 2004) and correlates with direct measurements of increasing atmospheric and ocean-water temperatures (causing ocean water to expand and increase its volume), and with observations of the melting of mountain glaciers and ice sheets (directly contributing water to the ocean), the primary factors important in causing a rise in sea level. By combining tide gauge analysis with satellite altimetry data, Church and White (2006) documented for the first time another statistically significant acceleration of the global SLR rate, this time in the 20th century, a confirmation of the climate change model simulations reported by the International Panel on Climate Change (IPCC).

Since the early 1990s, satellite altimetry has been used to measure sea level with approximately global coverage. This represents a significant advance over analyses of tide gage data, which are limited to continental coastlines and islands. Between 1993 and 2007 satellite altimetry data reveal a rate of global mean SLR of 3.3 ± 0.4 mm/year (Cazenave and Llovel, 2010). The global coverage of satellites has also exposed significant regional variability, with some areas experiencing as much as 3 times greater than the global rate during this 15 year period (e.g., the

western Pacific). The combination of altimetry, measurements of the mass of ice sheets and glaciers (via changes in the Earth's gravity field, gravimetry), and estimates of thermal expansion due to ocean warming allow for the development of a quantitative sea level budget during the period of 1993 to 2007. During that 15 year period it is estimated that approximately 1.0 ± 0.3 mm/year of the SLR rate was due to thermal expansion, 1.1 ± 0.25 mm/year was due to the melting of glaciers, and 0.7 ± 0.2 mm/year was due to the melting of ice sheets (0.4 ± 0.15 mm/year for Greenland and 0.3 ± 0.15 mm/year for West Antarctica). This sea level budget sums to 2.85 ± 0.35 mm/year and is approximately equal to the 3.3 ± 0.4 mm/year altimetry estimate considering the uncertainty in each of the measurements (Cazenave and Llovel, 2010).

Reports by the IPCC have projected that sea levels can be expected to rise at still higher rates during the rest of the 21st century in response to global warming that will increase ocean-water temperatures as well as accelerate rates of glacial melting (Bindoff et al., 2007). According to the recent IPCC projections (the fourth assessment report), the total increase in the average global sea level by the end of the 21st century will be significantly greater than the 15 to 20 centimeter rise during the 20th century. Projections range from 0.18 to 0.59 m of sea level rise by 2100.

It has been suggested, however, that even the more extreme IPCC projections might be on the low side since they did not account for the "wet processes" of glacial disintegration that are now evident for the Greenland and Antarctic Ice Sheets (Hansen, 2007). IPCC authors (e.g., Meehl et al., 2007) did suggest an additional 0.1 to 0.2 m of rise by 2100 based on future ice melt, acknowledging that the model projections could not account for the increased contributions due to ice melt because of the lack of scientifically based approaches at the time. In a recent attempt to constrain the range of possible sea level rise by 2100 with realistic physics of glacial melting, Pfeffer et al. (2008) suggested a maximum potential rise of 2.0 m by the end of the century, while reporting that estimates of ~0.8 m were probably more likely. Several other researchers have recently published semi-empirical projections of SLR, an approach that avoids modeling the complex glacier dynamics that the physics-based climate models are presently unable to accurately address. For example, Rahmstorf (2007) derived a semi-empirical relationship that connects global sea-level rise to global mean surface temperature and suggested that a projected sea-level rise in 2100 could be as much as 1.4 meters above the 1990 level. More recent semi-empirical estimates by several groups (e.g., Vermeer and Rahmstorf, 2009; Jevrejeva et al., 2010; Grinsted et al., 2009) suggest that the rise in sea level by 2100 may range between 0.59 to 2.15 meters. Limitations of these semi-empirical approaches include not being able to account for changes in nonlinear ice flows that are predicted to take place in the future (Rahmstorf, 2010). Therefore, these estimates might also be under predicting the amount of SLR that could occur by the end of the 21st century.

There clearly is a wide range of possible sea level rise scenarios for the 21st century, an indication of the significant uncertainty in our ability to make such predictions. Nonetheless, virtually all approaches predict a substantial increase in future sea level.

6.2.1 Relative Sea Level Rise in the U.S. Pacific Northwest

Studies of global SLR have, for the most part, sought to quantify spatially averaged changes in sea surface height due to the direct addition of water mass to the oceans as well as from thermal

expansion due to warming of the oceans. Superimposed upon the global signal are processes that cause regional redistribution of ocean volume, (e.g., regional variability in thermal expansion (Cazenave and Llovel, 2010)). The effects of these regional perturbations can be fairly large relative to the total water level change observed during the 20th century. While global SLR during the 20th century is estimated to have been approximately 1.7 mm/yr, Burgette et al. (2009), using several approaches, estimate that the regional rate of SLR has been approximately 2.3 mm/yr in the Pacific Northwest (PNW).

Komar et al. (in press) recently analyzed nine NOAA (NOS) tide gauges along the U.S. PNW coast (Figure 6.3) to document interdecadal trends and enhanced water levels that occur during strong El Niños. Hourly and monthly measured water level data were obtained from the NOAA's National Ocean Service, Center for Operational Oceanographic Products and Services[8] and analyzed using statistical approaches consistent with those of Holgate (2007) and Zervas (2009). The longest records along the U.S. PNW coast are those from Neah Bay, WA (75.4 years), Astoria, OR on the Columbia River (84.9 years), and the tide gauge at Crescent City, CA (77 years) near the Oregon-California border. The records from the other tide gauges range in lengths from 30 to 40 years, so the resulting analyses of their rates of changing sea levels are typically more uncertain.

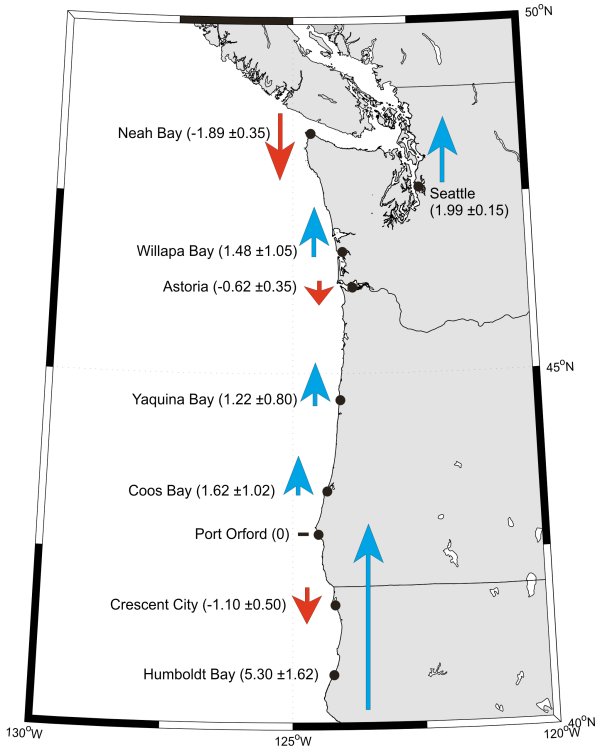


Figure 6.3 Analysis of NOAA tide gauges to assess changing sea levels along the coast of the PNW. Colored arrows represent the rates of change in relative sea levels (mm/yr), along with their uncertainty, generated using summer data only (see Section 6.2.2 for details). (After Komar et al., in press).

Examples of the analyses of the changing sea levels measured by two of the tide gauges are shown in the graphs of Figure 6.4, results for the Crescent City and Yaquina Bay tide gauges.

Each data point on the graph represents a mean sea level for that year, calculated by having averaged the hourly tide measurements over the entire year. It is evident that the trends of changing sea levels over the decades have been different at these two sites. The regression line through the Crescent City annual mean sea levels slopes downward at a rate of -0.78 mm/year (± 0.36 mm/year), signifying that the measured sea levels have been dropping at that site on the coast of northern California, whereas the curve for Yaquina Bay on the mid-Oregon coast shows that the measured sea level has been rising at a rate of 0.65 mm/year (± 1.33 mm/yr)[9]. While the Yaquina Bay annual average rate of rise is not significantly different from zero, due to the large amount of uncertainty associated with the calculated slope, there is a significant difference in the overall pattern of response between the two sites. This difference is caused by the tectonic induced changes in land elevations at these PNW coastal sites. In recognition that the rate measured by a tide gauge combines changes in both absolute (regional) sea levels and land elevations, the measured result is termed the relative sea level (RSL) change rate. In the case of Crescent City where there is a net drop in the RSL, the conclusion is that the land must be rising faster than the regional rise in mean sea level during the 20th century (2.3 mm/year). In contrast, the RSL rise measured by the Yaquina Bay tide gauge implies that at this mid-Oregon coastal site the extent of land-elevation change has been significantly less than has occurred at Crescent City.

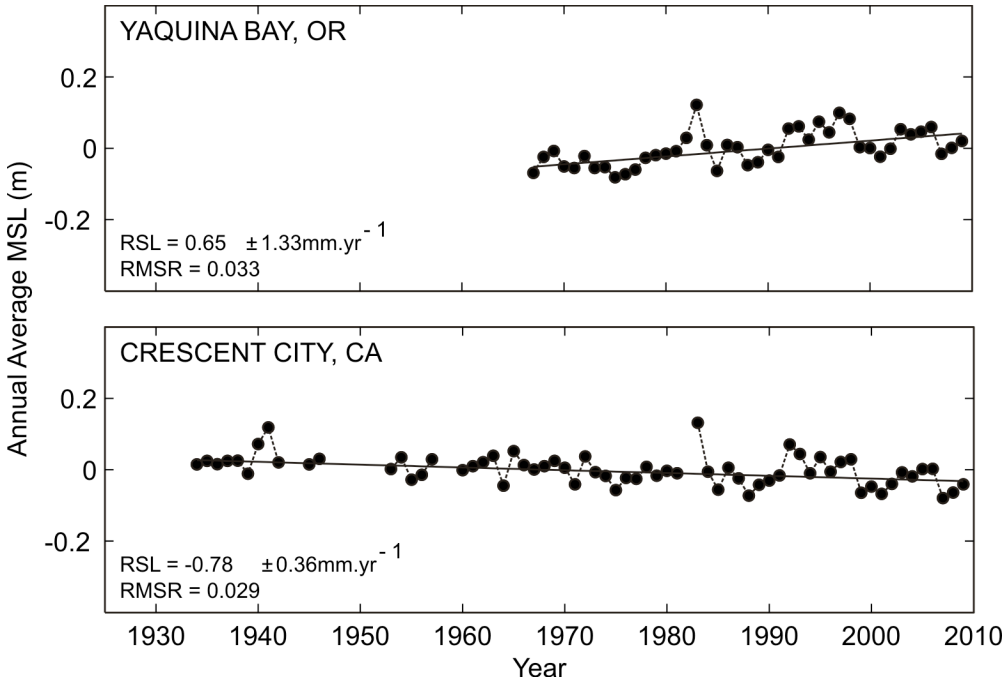


Figure 6.4. Analyses of the changing annual-average relative sea level (RSL) rise rates based on the tide-gauge records from Crescent City, California, and Yaquina Bay on the central Oregon coast. RMSR is the root mean square difference between the linear trend and the data. (After Komar et al., in press).

The land-elevation changes that have affected the measured sea levels at Crescent City and Yaquina Bay are due to the active tectonics of the PNW coast, involving the collision of Earth's tectonic plates, with the subduction of the oceanic plates beneath the continent (Figure 6.1). This plate subduction accounts for the greatest hazards faced along this coast, the potential for a massive subduction earthquake and tsunami like that which occurred on 26 January 1700

(Atwater et al., 2005). This tectonic setting is also altering the land elevations along the coast, in particular accounting for the stretches of shore that are rising at faster rates than the present-day rise in sea level. The magnitudes and directions in the rates of changing RSL determined from the PNW tide gauges are graphed as vectors in Figure 6.3, a red vector denoting that the tectonic uplift of the coast has been faster than the global rise in sea level, the measurements by the tide gauge being a drop in the RSL; the blue vectors represent results where there has been a trend of rising RSLs. There is seen to be signficant variations in the RSLs along the coast, reflecting the tectonic control on the rates of land elevation changes.

Geophysicists have been able to directly assess the changes in land elevations based on models and periodic resurveys of established bench marks used by surveyors and their results are in reasonably good agreement with those derived from the tide-gauge records (e.g., Burgette et al., 2009; Mazzoti et al., 2008; Verdonck et al., 2007, Mitchell et al., 1994). The along coast patterns in RSL are developed in greater detail in Figure 6.5, where the alongshore varying RSL changes measured by the tide gauges are compared with both the direct measurements of the changes in land elevations derived from resurveys of benchmarks and land elevations measured at GPS sites along the coast (Komar et al., in press). To make the comparisons in Figure 6.5, it was necessary to subtract the rates of land elevation changes derived from the benchmarks and GPS measurements from the magnitude of the regional rise in sea level, 2.3 mm/year, as determined by Burgette et al. (2009), yielding the alongshore patterns in RSL rates graphed in Figure 6.5.

The different rates of land elevation changes along the coast and the resulting variable trends in the RSL rates evident in Figure 6.5 represent a primary control on the corresponding along-coast variations in property erosion and flooding impacts. It is apparent from the geomorphology of the coast and locations of communities which have experienced erosion, that the stretches of shore that are tectonically rising faster than the global rise in sea level (such as Crescent City) have been relatively immune from those hazards, while the areas that are not rising rapidly (such as in Tillamook County, Oregon) are those that have experienced the greatest impacts from erosion and flooding. The application of SLR estimates in decision making will depend on location, time frame, and risk tolerance. For decisions with long timelines and low risk tolerance, such as coastal development and public infrastructure, users should consider low-probability high-impact estimates that take into account, among other things, the potential for higher rates of SLR driven by recent observations of rapid ice loss in Greenland and Antarctica, which though observed were not factored into the IPCC's latest global SLR estimates. Combining the IPCC high emissions scenario with 1) higher estimates of ice loss from Greenland and Antarctica, 2) seasonal changes in atmospheric circulation in the Pacific, and 3) vertical land deformation, a low-probability high-impact estimate of local SLR for the areas of the coast experiencing little vertical land motions is 55 cm (22") by 2050 and 128 cm (50") by 2100. However, by the mid 21st century the projected increase in rates of SLR are expected to exceed the rates of uplift of the land all along the Oregon coast, resulting in erosion even where at present there has been little or no erosion impacts. The scenario as to when this enhanced erosion and flooding begins at a specific coastal site, and the magnitude of that enhancement, depends however on the contributions by other oceanic processes and their climate controls, particularly the increase in storm intensities and the heights of their generated waves (Ruggiero, 2008).

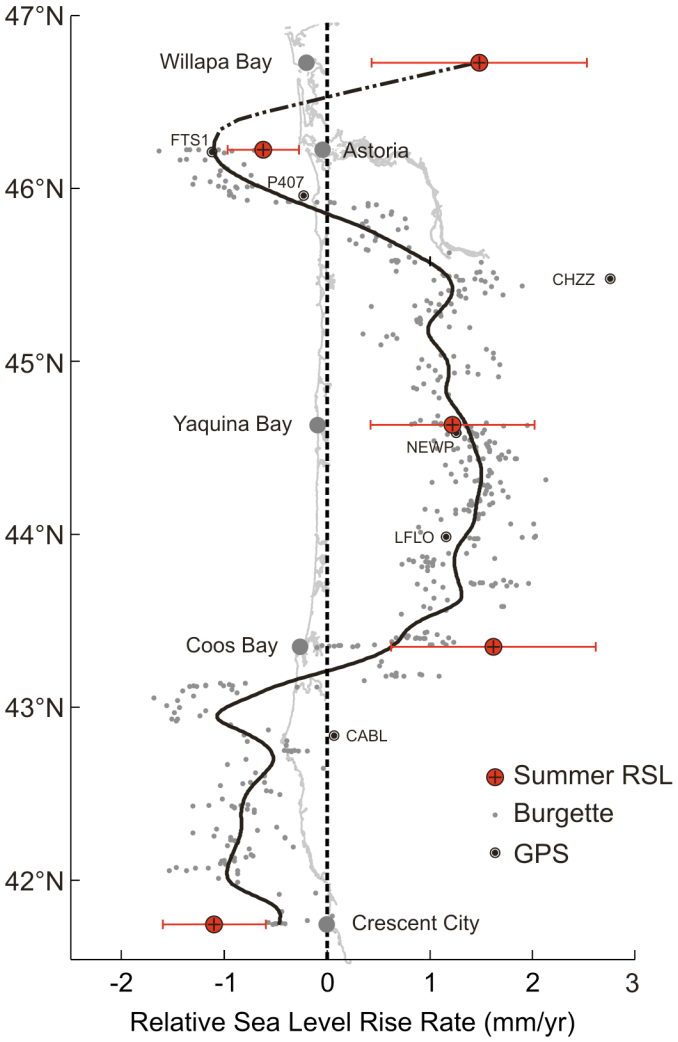


Figure 6.5 Alongshore varying rates of relative sea level (RSL) rise as determined by three methods. 1) Tide-gauge records with trends based on averages of the summer only monthly-mean water levels (red circles with plusses, error bars represent the 95% confidence interval on the trends). 2) Subtracting the Burgette et al. (2009) benchmark survey estimates of uplift rates from the regional mean SLR rate (2.3 mm/yr) (small gray dots). 3) Subtracting the uplift rates estimated from GPS sites along the coast from the regional mean sea level rate (small filled black circles). [From Komar et al. (in press)].

6.2.2 El Niño Induced Water Levels

While the interdecadal trends of changing sea levels and their future increasing rates pose a threat to the Oregon coast, it is the year-to-year variations that produce the extreme water-level elevations that cause episodes of erosion and flooding. This variability is evident in the curves of Figure 6.4 for the annual mean sea levels measured by the Crescent City and Yaquina Bay tide gauges, the peaks having been caused mainly by El Niños, the most conspicuous having been those for the 1982-83 and 1997-98 major El Niño climate events. A strong El Niño typically raises the monthly-mean water levels by 10s of centimeters, and therefore during extreme El Niños winter monthly mean water levels can exceed the magnitude of the projected global rise in sea level expected during the next few decades. The resulting impacts from erosion and flooding

along the Oregon coast have led to significant damage and losses of coastal properties (Komar 1997, 1998; Allan et al., 2003; Allan and Hart 2008; Allan et al. 2009).

The full effects of El Niños on the mean water levels during the winter are seen in Figure 6.6. The figure shows monthly means based on analyses of the tide-gauge record from Yaquina Bay. The records from other gauges along the PNW coast are closely similar in their magnitudes and the timing of the seasonal variations. The curve in this diagram for the long-term (1967-2009) average cycle shows that the highest monthly-mean water levels always occur during the winter months of December through February, the lowest being in the summer, with the normal difference between the two seasons being on the order of 0.25 m. This occurrence of the highest water levels during the winter is unusual in that on most coasts the highest levels occur in the summer when the water is warmest, resulting from solar radiation and the heating and thermal expansion of the water. However, along the coast of the PNW the development of upwelling during the summer (Chapter 1) brings cold dense water up to the surface; the water is therefore coldest in the summer and warmest during the winter. This unusual cycle of water temperatures along the PNW coast, together with the changing directions and magnitudes of the ocean's currents, produce the seasonal changes in monthly-mean water levels seen in Figure 6.6. This represents an important factor in occurrences of coastal erosion and flooding since the highest water levels occur at the same time as the strongest storms and their extreme waves.

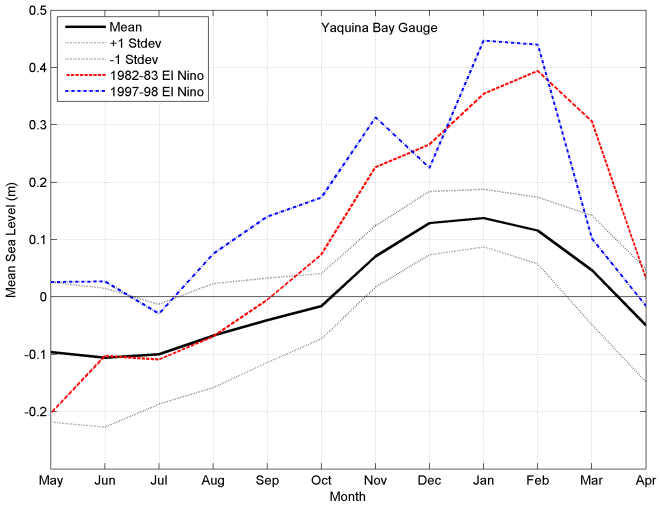


Figure 6.6 Seasonal cycles in the monthly mean water levels derived from analyses of the Yaquina Bay tide-gauge data, including the long-term averages and the cycles for the 1982-83 and 1997-98 El Niño years. (After Allan and Komar, 2002).

Also included in Figure 6.6 are the extreme seasonal cycles that occurred during the strong 1982-83 and 1997-98 El Niños. It is seen from their cycles that the monthly-mean water levels in the winter were some 50 centimeters higher than during the previous summer, approximately double the normal seasonal cycle. During the months of November through March the monthly-mean water levels of these strong El Niños were on average about 30 centimeters higher than occur during the normal long-term average cycle. The effect was much like there having been a 10- to 30-centimeter rise in sea level over a span of about 6 months; the average rate of rise would have been equivalent to 600 mm/year, about 300 times the rate of the global rise in sea level

spanning the 20th century. Fortunately, when each of these El Niños came to an end their elevated water levels returned to normal, but during those several months of high water levels considerable erosion, and occasional flooding, occurred along the Oregon coast.

The implications of this magnitude of increased water levels during El Niños to erosion and flooding is obvious, since it would have raised the levels of all tides by that amount, the low tides as well as high tides. This was particularly evident during the strongest El Niños in that many of the beaches were "flooded out" at all stages of the tides, there having been almost no dry beach throughout the winter. And at times of high tides, the elevated water levels were able to reach the shore-front properties, the toe of sea cliffs or foredunes backing the beach, resulting in significantly more erosion than occurs during normal (non-El Niño) winters.

It is apparent that episodes of extreme water levels during major El Niños need to be accounted for in evaluating coastal hazards and their climate controls. Some researchers have suggested that global warming may affect the frequency and intensities of El Niños (e.g., Trenberth and Hoar, 1997), however, at present there is no consensus about whether their activity will be enhanced or decrease in the future with a warmer Earth. The analysis in Figure 6.7 is much the same as presented in Figure 6.4 to examine the trends in the annual mean sea levels, but here the monthly-mean water levels for the winter and summer are separately analyzed, the objective being that by limiting the analysis to the winter months the result will emphasize the extreme water levels during El Niños (Komar et al., in press). To make the results particularly meaningful to hazard assessments, "winter" is defined as the combined average tide levels measured over a 3-month period around the peak of the seasonal maximum in water levels, typically the months of December through February. Similarly, "summer" represents the average tide level measured over a 3-month period around the seasonal minimum, typically the months between May through July when levels also tend to be least variable.

The contrast between the "winter" and "summer" seasons is evident in Figure 6.7, there being more variability in the winter record, with the peaks for the major El Niños being most prominent. In order to provide meaningful assessments of the contributions by El Niños to the measured water levels during the winter, it is necessary to detrend the data to remove the effects of the land elevation changes and the regional rise in sea level (Allan and Komar, 2006; Komar et al., in press). The resulting detrended "residual" winter water levels for each year of the Yaquina Bay record (not shown) have been correlated with the corresponding annual values of the Multivariate ENSO Index (MEI), which climatologists have developed to provide a measure of the ENSO events ranging from La Niñas to El Niños. The results document a well-defined increase in the residual

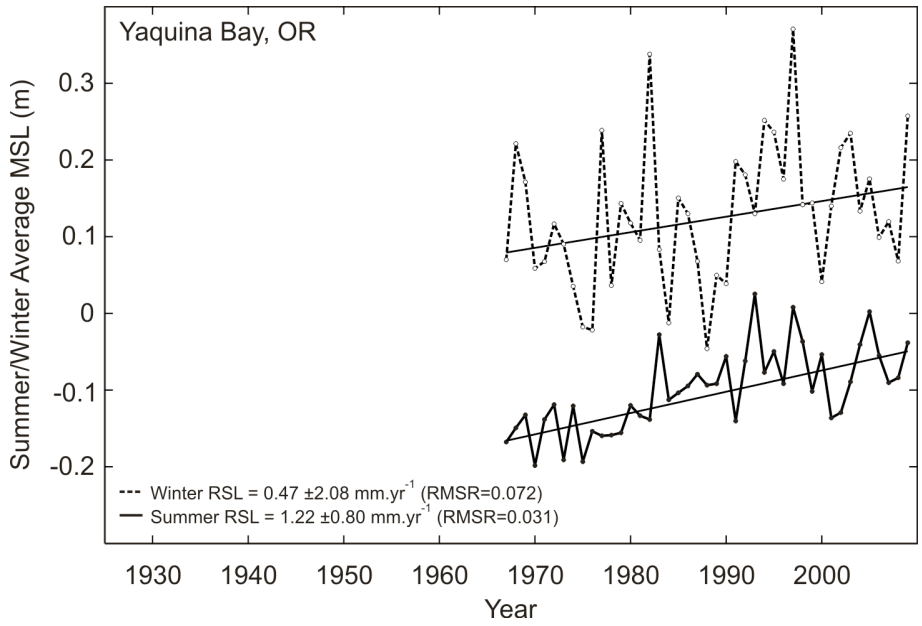


Figure 6.7. Mean sea levels for the "winter" and "summer" based on the Yaquina Bay tide gauge. Relative sea level (RSL) rise rates are given along with the uncertainty at the 95% confidence interval. The root mean square residual (RMSR) difference between the linear fit and the data are also given. [From Komar et al. (in press)]

water levels with increasing Multivariate ENSO Index, confirming that the highest elevated water levels occur during the strongest El Niños, while La Niñas tend to depress water levels. Closely similar correlations were found for the records based on the other PNW tide gauges, confirming the existence of a fairly uniform along coast response of the elevated winter water levels to occurrences of El Niños.

Figure 6.7 illustrates that the data and linear regression for the summer months are systematically displaced about 0.25 m below the results for the winter, this difference reflecting the seasonal change in mean water levels earlier seen in Figure 6.6. Of significance, the variability caused by El Niños has been almost entirely removed in this graph for the summer months, reducing the scatter so that the long-term trend is better defined. The result is that the analysis based on the summer data only provides a better assessment of the decadal trend in RSL than the annual averages and therefore, only the summer-average derived trends have been included in Figures 6.3 and 6.5.

This series of analyses of the records derived from the tide gauges along the PNW coast have documented their interdecadal trends of increasing or decreasing RSLs, with the direction of change being dependent on the tectonic control on the rate of uplift of the land (Komar et al., in press). Superimposed on the resulting trend in the RSLs are the additive extremes of elevated winter water levels during major El Niños. As rates of global sea level rise accelerate during the 21st century, it can be expected that the "winter" average water levels together with El Niño-induced extremes will progressively shift to higher elevations. This pattern of increasing sea-level extremes will result in greater impacts from erosion and flooding along the Oregon coast.

6.3 Changes to Storm Climatology

The Oregon coast is well known for the severity of its winter storms and the heights of the waves they generate. During storms, the deep-water significant wave heights are regularly greater than 10 meters (about 1 storm of this size per year), the "significant wave height" being defined as the average of the highest one-third of the measured wave heights within a (typically) 20-minute period. Being something of an average, there are larger individual waves generated by the storm, with the maximum height being approximately a factor 1.7 greater than the significant wave height; therefore, for a storm with a 10-meter significant wave height, individual waves having heights up to ~17 meters can be expected. The most extreme storm in recent years in terms of the heights of the waves measured by offshore buoys occurred in early March 1999, when the significant wave heights reached 14 to 15 meters (Allan and Komar, 2002). With that 1.7 factor, the highest individual waves during that storm would have been ~25 meters, the height of a 10-story building. Being capable of generating such extreme waves, Oregon's winter storms have been the dominant factor responsible for episodes of erosion and flooding along this coast (Ruggiero, 2008).

The first buoys designed to measure waves off of Oregon's coast were deployed by NOAA in the mid-1970s, providing hourly measurements of wave heights and periods. Of concern, the heights of these waves have been increasing over the decades (Allan and Komar 2000, 2006; Ruggiero et al., 2010), a result of changing extratropical storm characteristics.

6.3.1 Extratropical Storm Intensity, Frequency and Tracks

Allan and Komar (2000, 2006) analyzed averages of winter significant wave heights and documented that, since the 1970s when wave buoys became operational, wave heights had increased off the shores of Washington and Oregon. Lower rates of increase were observed off the northern and central California coasts, while waves off southern California were found to have experienced little net change. Analyses by climatologists of North Pacific extra-tropical storms have concluded that their intensities (wind velocities and atmospheric pressures) have increased since the late 1940s (Graham and Diaz, 2001), implying that the trends of increasing wave heights likely began in the mid-20th century, earlier than could be documented with the direct measurements of the waves by buoys. Graham and Diaz (2001) suggested that the intensification of North Pacific winter storms has resulted from increasing upper-level winds; a finding that had been observed earlier by Ward and Hoskins (1996). Graham and Diaz (2001) further hypothesized that this increasing trend in upper-level winds might be the result of global warming, specifically the increased sea surface temperatures in the western tropical Pacific. An alternative hypothesis has suggested that "black carbon" aerosols (pollution) derived from industries in India and China could also be contributing to the increase in storm intensities (Zhang et al., 2007).

Additional research on trends in mid-latitude extra-tropical storms in the Eastern North Pacific has confirmed that there has been an increase in intensity, but a decrease in frequency, possibly since the storm tracks have shifted poleward during the latter half of the 20th century. McCabe et

al. (2001) showed a statistically significant decrease in the frequency of storms over the years 1959-1997. Geng and Sugi (2003) found that the decrease in annual numbers of storms is typically of the weak-medium strength variety, while the stronger storms have actually increased in frequency.

These documented changes in storm tracks are thought to be primarily due to changes in temperature and pressure gradients, which in turn are linked to changes in atmospheric temperature distributions due to increased greenhouse gas emissions. In other words, in the mid-latitudes of the Northern Hemisphere a decrease in the meridional temperature gradient (poles are warming faster than lower latitudes) has led to a decrease in mid-latitude storm frequency. Recognizing the trends in reanalysis data, Yin (2005) used the output of 15 coupled general circulation models to relate the poleward shift of the storm track to changes in baroclinicity in the 21st century. Though these studies were conclusive that the storm track shifts poleward in the Northern Hemisphere with warmer temperatures, uncertainties regarding natural variability and model limitations remain.

Recently, Favre and Gershunov (2006) analyzed wintertime cyclones (low pressure "storms") and anticyclones (high pressure "calms") over the Northeast Pacific for the period 1950 to 2001. They observed that while the strength of anticyclones had gradually diminished over the period and their frequency had become more variable, extratropical cyclones had intensified (consistent with the earlier work of Graham and Diaz (2001)) in a discrete shift with deeper more intense lows and further southerly trajectories occurring since the mid-1970s. This latter response produces a dipole of sea level pressure with an anomalously deep Aleutian Low and higher pressure in the Western Pacific inter-tropical zone, characteristic of the prevailing ENSO conditions and warm phase of the Pacific Decadal Oscillation that have dominated the North Pacific over the past three decades. However, the exact cause of these changes and the degree of intensity was not explained.

6.3.2 Increasing Wave Heights

Of particular significance to Oregon's coastal hazards is that the more extreme waves generated by the strongest storms are increasing at appreciably higher rates than are the "winter" averages (Allan and Komar, 2006; Ruggiero et al., 2010). This is shown in Figure 6.8, with the series of data plots and linear regressions ranging from the annual average measured significant wave heights, the winter averages, the average of the 5 largest measured significant wave heights that occurred each winter, and the annual maximum significant wave height representing the most severe storm each year. Apparent from this series in Figure 6.8 is that the more extreme the wave statistic analyzed, the steeper the slope of the linear regressions that represent the rates of increasing significant wave heights over the decades. While the averages of all significant wave heights measured during the winter have been increasing at a rate of 0.023 m/year, the maximum significant wave heights of the strongest storms have been increasing at the substantially higher rate of 0.095 m/year. As given by the regression line, this maximum has increased from about 9 meters in the latter 1970s to about 12 meters in 2005, the ~30-year span of measurements from the NOAA buoys.

Another meaningful depiction of the changing wave climate is provided by histograms for the full range of measured significant wave heights (Komar and Allan, 2007). Figure 6.9 compares a pair of frequency counts based on wave data derived from the same NOAA buoy as analyzed in Figure 6.8, the separate curves comparing the ranges and numbers of significant wave heights measured during the 1976-1989 versus 1996-2006 "decades" (intervals during which nearly identical numbers of measurements were obtained). The shift to higher extreme measured significant wave heights is apparent in this pair of histograms. There is also a suggestion that the lowest measured significant wave heights, those that occur during the summer, have also experienced an increase over the decades (Ruggiero et al., 2010).

Ruggiero et al. (2010) pointed out that since the wave height climates are governed by the log-normal probability distribution, a modest increase in the annual mean wave height can result in a significant impact on the frequency and magnitude of extreme events. Therefore, measurements of significant wave heights off the U.S. PNW coast represent a clear example of a phenomenon that was suggested by Wigley (2009) in general terms; a gradual change in the mean climate of an environmental variable can result in significant increases in the frequency of extreme events.

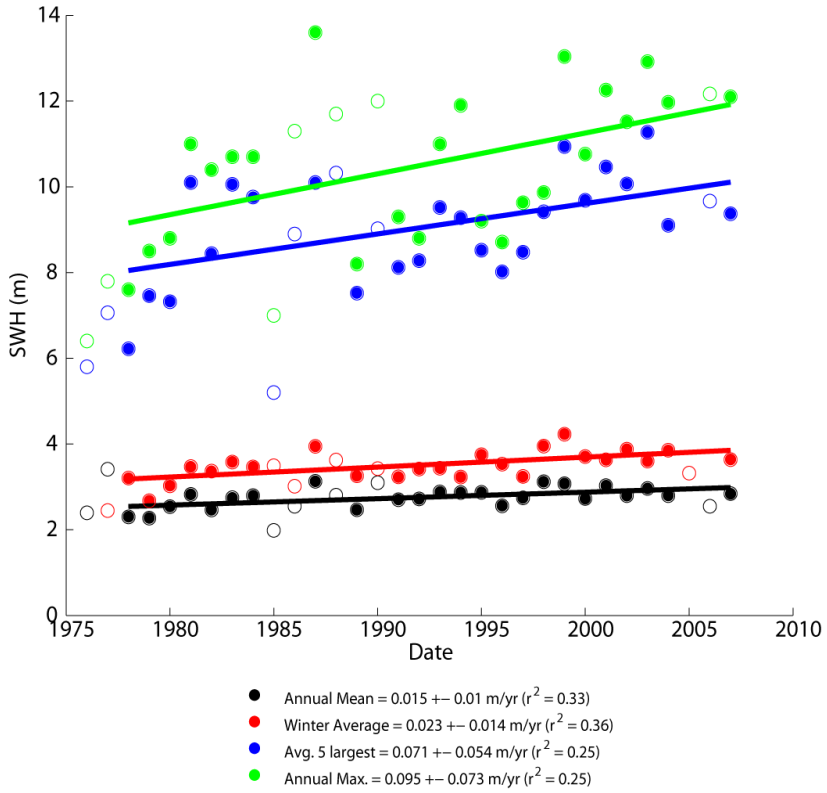


Figure 6.8 Decadal increases in annual mean, winter average, average of the five largest per year, and annual maxima SWHs measured by NDBC buoy #46005 off the PNW coast. The regression slopes and the uncertainty of the regression slopes are given along with the r^2 values. Each of the regressions is statistically significant at the 95% confidence level. Open circles represent years that did not satisfy the criterion for inclusion in the regressions (after Ruggiero et al., 2010) and have not been included in the regressions.

Taken together the analyses in Figure 6.8 of the decadal trends in the average significant wave heights and in Figure 6.9 of the shift in the number distributions provide documentations of the increasing storm-generated waves measured by buoys off the coast of the PNW. To a degree they can provide guidance as to the magnitudes of increases in extreme-value projections, for example the 100-year events that are needed by coastal decision makers in coastal-hazard assessments and by coastal engineers in their designs of coastal structures. However, formal statistical analysis procedures have been developed that can be applied to time-varying changes in data populations, with many directed toward the environmental consequences of global warming (e.g., temperatures, rainfall, and river discharges). Such statistical procedures represent a significant advance over classical extreme-value theory, and have been applied in analyses of the decadal changes in the wave climate of the PNW. Ruggiero et al. (2010) showed that the 100-year wave height, for NDBC Buoy #46005, has been increasing at a rate of approximately 0.07 m/year, a result that is in good agreement with the linear regression for the highest measured significant wave heights (Figure 6.8). By having analyzed the extreme values using the more advanced procedures, the statistical significance and confidence in the results has been considerably improved (see Ruggiero et al., 2010, for details).

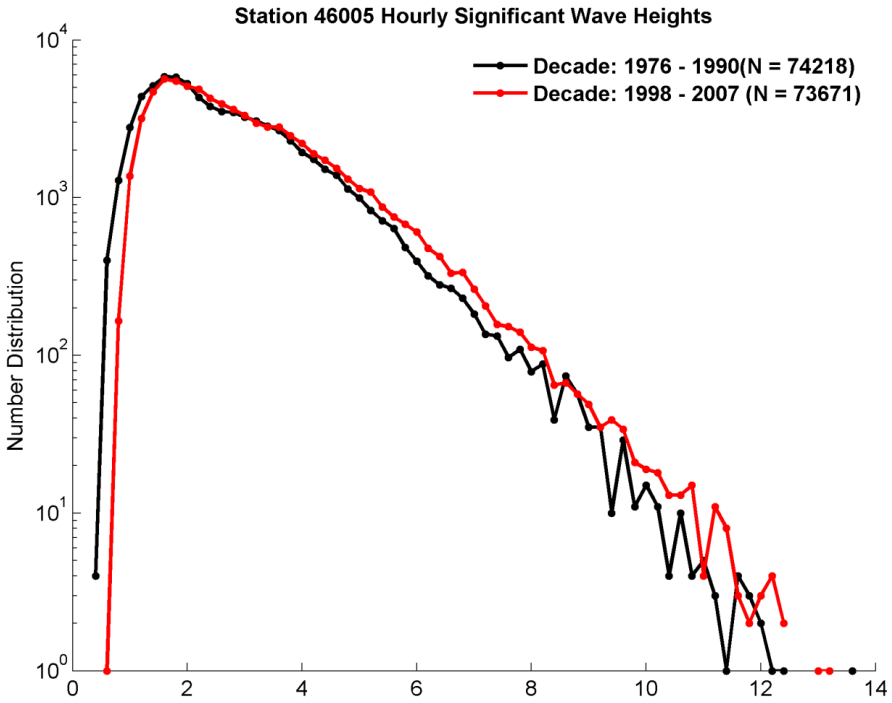


Figure 6.9 Number distributions documenting changes in the ranges of measured significant wave heights off the coast of Washington (modified from Ruggiero et al., 2010).

The wave conditions along the U.S. west coast are also altered by the occurrence of strong El Niños. Most significantly, the coast of southern California is impacted by unusually high waves due to the altered tracks of the storms, so they cross the coast of central to southern California

during an El Niño, rather than following their more normal tracks that take them across the shores of the PNW. Although the storms are more distant from the PNW coast during an El Niño, their generated waves that reach the Oregon coast have an important role in the resulting erosion impacts. Most significant are the altered directions of the waves that arrive more from the southwest than normal. This dominance of the waves reaching the Oregon coast from the south to southwest produces a redistribution of the sand on its beaches, creating what is termed "hot spot" erosion that is characteristic of El Niño winters.

6.4 Climate Change Impacts on Coastal Hazards

It is evident that coastal change and flood hazards along the Oregon coast are caused by a number of ocean processes, each of which has significant climate controls such that the severity of the problems in the future can be expected to increase. Important is the near certainty that the rate of sea-level rise will be greater in the future as a result of global warming. Evaluating the consequences to enhanced erosion and flooding is complicated by Oregon's tectonic setting, with there being significantly different rates of uplift along the coast. Taken together, the variable uplift of the land plus the present-day rate of regional SLR, stretches of the coast are submerging whereas other areas are emerging. The prospects are that with the expected accelerated rates of SLR, the entire coast will at some point be submerging and experience significantly greater erosion and flooding impacts than at present day. Another long-term trend important to the future coastal change and flood hazards along the Oregon coast, possibly linked to global warming (Section 6.3.1), is the increasing intensity of major winter storms and the heights of the waves they generate. Application of simple coastal hazards models indicate that over the period of wave-buoy observations (~30 years) wave height (and period) increases have had a more significant role in the increased frequency of coastal flooding and erosion in the PNW than has the rise in sea level (Ruggiero et al., 2008). In addition to those progressive trends, the periodic occurrence of major El Niños in the future will compound the impacts of increasing sea levels and waves, resulting in the most severe episodes of Oregon coast erosion and flooding. With these multiple processes and their climate controls having important roles in causing erosion and flooding along the Oregon coast, it is challenging to collectively analyze them with the goal of providing meaningful assessments of future hazards during the next 25 to 100 years.

6.4.1 Observations of Coastal Change along the Oregon Coast

To understand the impacts of storms, particularly during major El Niños, and to improve our understanding coastal change due to climate change, staff from the Newport Coastal Field Office of the Oregon Department of Geology and Mineral Industries (DOGAMI), have developed the "Oregon Beach and Shoreline Mapping and Analysis Program (OBSMAP)[10]". The OBSMAP program has two broad goals: first, to provide up-to-date information on the state of beaches and shorelines along the PNW coast, information that can be used by coastal managers, geotechnical consultants, and the public-at-large, and second to develop an improved understanding of the seasonal-interannual-decadal changes in Oregon beaches. Beach monitoring undertaken as part of the OBSMAP effort is based on repeated high-accuracy GPS surveys of selected beach profiles (e.g. Ruggiero et al., 2005; Allan and Hart, 2008). The OBSMAP monitoring network currently consists of 119 active beach monitoring sites that are funded through the Northwest Association

of Networked Observing Systems (NANOOS), and another 85 sites that have been observed on a case-by-case basis.

6.4.1.1 El Niño induced coastal change

The occurrence of concentrated "hot spot" erosion during major El Niño events is due to the fact that the Oregon coast consists of a series of littoral cells, stretches of beach that are confined by bounding rocky headlands (Komar, 1997). The significance is that as diagramed in Figure 6.10; during normal years there typically is a seasonal reversal in the longshore movement of the sand within each littoral cell, transported to the north during the winter by waves that arrive from the southwest, but during the summer with the waves arriving principally from the northwest an approximately equal volume of sand moves back to the south. The shoreline thereby maintains a nearly balanced equilibrium in the north-south oscillations of its sand during normal years. However, as also diagramed in Figure 6.10, during an El Niño with a greater dominance of waves arriving from the southwest in the winter, a greater volume of sand is transported to the north within Oregon's littoral cells. The consequence of this northward displacement of the beach sand within the cells is that the beaches at their south ends, to the north of the headlands, undergo massive erosion, becoming the primary zones of hot-spot erosion during El Niños (Komar, 1986, 1998; Peterson et al., 1990; Kaminsky et al., 1998; Revell et al., 2002; Allan et al., 2003; Allan et al., 2009). Erosion also tends to occur to the north of inlets into bays and the mouths of rivers, when they migrate to the north under the forces of the waves arriving from the southwest, or to the north of jetties that can act like mini-headlands.

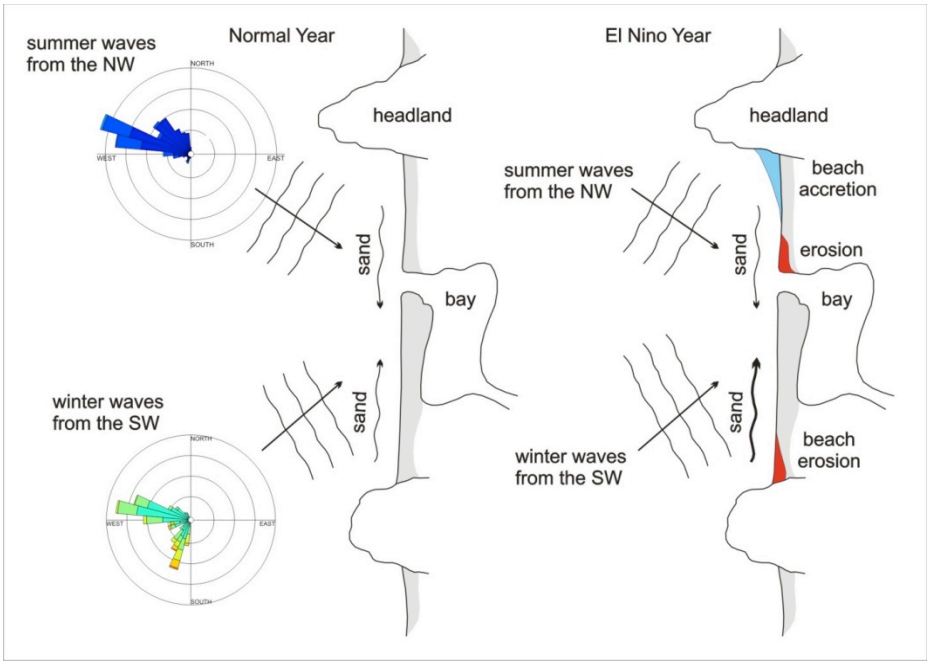


Figure 6.10 Sand movement along the beaches within Oregon's littoral cells between rocky headlands due to the seasonal changes in wave directions, contrasting the equilibrium balance during normal years with that during a major El Niño when the waves transport greater volumes of sand to the north, resulting in zones of "hot-spot" erosion (after Komar et al., 2002).

The hot-spot zones of erosion during the major El Niños of 1982-83 and 1997-98 represent some of the most significant impacts to coastal properties in recent decades. That erosion was the

combination of the exceptionally high water levels experienced during the winter months, and the northward movement of the beach sand by the waves that reach the coast from the southwest, so that the property losses were greatest in the hot-spot areas. Examples of significant hot-spot erosion problems along the Oregon coast include the following.

•Neskowin, with the hot-spot area of maximum beach and foredune erosion having occurred immediately north of Cascade Head.

•The erosion and flooding impacts to Cape Lookout State Park at the south end of Netarts Spit, to the north of the Cape, during both the 1982-83 and 1997-98 El Niños.

• Impacts to The Capes development of condominiums that were constructed on a high sand bluff that was eroded by the northward migration of the inlet to Netarts Bay.

•Extensive erosion of the Bayshore development on Alsea Spit during both major El Niños, caused by the northward migration of the Bay's inlet.

•The erosion of the beach and foredunes in Port Orford north of The Heads, resulting in the loss of the community's sewage disposal facility, and leading subsequently to a breach through the dunes that carried water into Garrison Lake that was its source of fresh water.

It is clear that the extreme processes and climate controls of El Niños need to be accounted for in hazard assessments along the Oregon coast, in addition to the climate controls on increasing sea levels and storm-generated wave heights. While it has been suggested that global warming may affect the frequency and intensities of El Niños (e.g., Trenberth and Hoar, 1997), at present scientists cannot definitively say whether their activity will be enhanced or decrease in the future with a warmer Earth (Collins et al., 2010).

6.4.1.2 Interannual- to decadal-scale coastal change

It is typical of the examples of erosion that began during one (or both) of the major El Niños that the erosion can continue for several years after the El Niño winter has ended. Furthermore, in the case of the 1997-98 El Niño it was followed by the winter of 1998-99 during which there was a series of unusually severe storms and extreme waves, including that in early March 1999 which generated the 14- to 15-meter significant wave heights, the highest in recent decades (Allan and Komar, 2002). These two years of consecutive impacts were characterized as having been a "one-two punch" that was responsible for extensive continued erosion. To-date many of the beaches on the northern Oregon coast, particularly in Tillamook County, have yet to fully recover from the cumulative effects of the 1997-98 El Niño and the more extreme 1998-99 winter (Allan and Hart, 2008). As a result, at the time of writing this report (summer/fall 2010) a number of beaches remain in a degraded state making them highly susceptible to potentially catastrophic future events that will inevitably take place during the next 'big' winter.

Erosion has become particularly acute along the Rockaway littoral cell (Figure 6.2) in Tillamook County, much of which can be attributed to the extreme storms that impacted this section of the coast during the 1997-1998 and 1998-1999 winters. Figure 6.11 was derived by analyzing topographic beach volume changes collected using airborne lidar (light detection and ranging)

data flown in 1997 and 2002. Subsequent monitoring of the beaches indicates that the shoreline has continued to experience ongoing erosion, with the Rockaway sub-cell having continued to lose sand over time, while Bayocean Spit and portions of the Nehalem Spit appear to be slowly gaining some of this sand; the net loss of sand from the cell is now estimated to be approximately two to two and a half million cubic meters of sand (Allan and Hart, 2008). Portions of the shore immediately north of Tillamook Bay have eroded by as much as 47 m, increasing the hazard potential to existing homes and infrastructure from ocean flooding and additional future shoreline retreat (Figure 6.12).

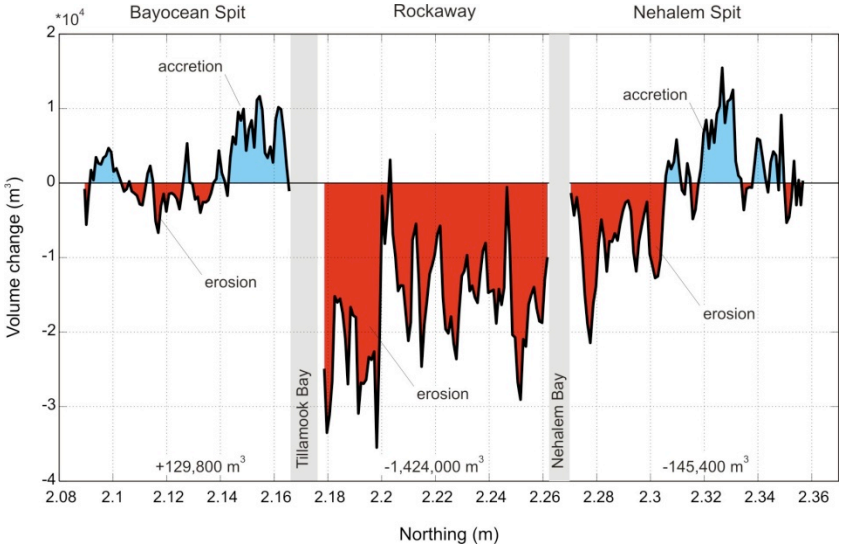


Figure 6.11 Alongshore beach volume changes (in cubic meters) derived from an analysis of available lidar data for the period 1997–2002. Red shading indicates erosion, blue shading indicates accretion.

Findings from the OBSMAP beach monitoring program reinforce two important points concerning PNW beach processes. First, high ocean waves exert a primary control on PNW beach and shoreline responses. Second, several littoral cells remain in a state of sediment deficit due to the volume of sand that has been removed from the beaches and dunes over the past decade. This is somewhat surprising since beaches typically begin to rebuild following major storms, and especially during the summer months when wave energy levels are significantly lower. However, analyses of the offshore wave climate as noted previously indicates that in fact wave energy levels and hence the wave heights have remained high over the past decade, and wave heights have been increasing since at least the mid 1970s (Allan and Komar, 2006). As a result, many Oregon beaches have continued to erode over the past decade, with signs of recovery having been confined to only a few discrete locations (e.g., the northern end of the littoral cells and sub-cells).



Figure 6.12 Ongoing shoreline retreat over the past decade in the Rockaway cell and localized hotspot erosion effects have resulted in substantial sections of the shore having to be rip-rapped in order to safeguard property. SLR expected over the next century and enhanced storms will almost certainly increase the risk of failure of such structures and the potential loss of homes and important infrastructure backing the beach.(Photo courtesy of Mr. Don Best, 2009.)

6.4.1.3 Long-term coastal change

Relatively few studies have attempted to quantify the long-term (decadal to century) patterns of shoreline change on the Oregon coast. Of the studies that have documented long-term coastal change, most have primarily focused on bluff changes in varying geologic units on the central Oregon coast (e.g. Priest, 1999; Allan and Priest, 2001; Priest and Allan, 2004; Witter, Allan and Priest, 2007). Estimates of long-term erosion rates in these studies were typically determined by comparing changes along the bluff top or toe at specific locations where ground-control points could be identified using aerial photographs. Using this approach, Priest (1999) found mean erosion rates to be low, averaging ~0.19 m/yr in northern Lincoln County, but could reach as much as 0.5 m/yr depending on the local geology. In contrast, Witter et al. (2007) identified erosion rates in south Lincoln county that were generally much lower (~0.03-0.15 m/yr).

For dune-backed beaches, Allan et al. (2003) examined long-term changes at several sites on the Oregon coast, including most of Tillamook County and Port Orford on the southern Oregon coast. They concluded that the largest measurable net shoreline changes were those that resulted from jetty construction. Elsewhere, Allan et al. (2003) demonstrated that the Oregon coast is subject to large temporal and spatial variations in its shorelines, varying at any one site from extreme erosion to accretion. These changes are a response to the high-energy character of Oregon's coastal processes, to occurrences of extreme storms, and particularly to major El Niños. Therefore, Allan et al. (2003) concluded that historical shoreline change assessments based on aerial photographs and NOS "T" sheets were generally not appropriate for making projections of future shoreline positions.

6.4.2 Projecting future hazards

Quantifying the impact of climate change on coasts is not straightforward because the response of these systems is a complex morphodynamic issue. In the absence of significant sediment sources and sinks, the Bruun Rule (Bruun, 1962) conceptually describes the retreat of unprotected coastlines due to SLR (via mass conservation). While the Bruun Rule is often pilloried for its simplicity, Stive et al. (2009) reviewed the various attempts to verify the approach against field and laboratory data over the last 4 decades and concluded that the basic concept of the Bruun Rule has been qualitatively confirmed. For many of the world's coastlines, where the nearshore beach slope is about 0.01 to 0.02, the Bruun Rule predicts a coastline retreat between 50xSLR and 100xSLR, values that are commonly used as a "rule-of-thumb". Unfortunately, the applicability of the Bruun Rule is usually compromised, particularly in the vicinity of tidal basins (Oregon has 43 estuaries.) Because of this, Stive et al. (2009) recommend that, at best, any predictions obtained via the Bruun Rule should be considered only as broadly indicative, order-of-magnitude estimates. Stive et al. (2009) go on to suggest that comprehensive bottom-up (small-scale, process-based) and top-down (large-scale, behavior-based) numerical modeling, validated by field data, is necessary to provide more scientifically robust and reliable predictions of coastline retreat due to future SLR. Unfortunately, this approach requires significant resources and is still under development, therefore it has yet to be applied to the Oregon coast.

One approach that is being followed to develop hazard zones along the Oregon coast (e.g., Allan and Priest, 2001; Baron et al., 2010) is the application of a series of simple models, with the first directed toward combining multiple processes to calculate the total water levels (TWL) at the shore and the frequency with which they can reach and impact shore-front properties, followed by simple models developed to evaluate the resulting property erosion (e.g., Komar et al., 1999). The first in the series of analyses is to calculate the 'nearshore processes climate' from the measured deep-water wave heights and periods. Next is the application of a simple TWL model (e.g., Ruggiero et al., 2001) which combines the varying levels of the tides with the runup levels of the storm waves to determine the TWLs achieved on beaches. Here the measured tides are used rather than the predicted astronomical tides, so as to include processes such as storm surge and the elevated winter water levels that would occur during a major El Niño. In making long-term projections, trends of increasing wave heights and future SLR are incorporated in a scenario-based approach. The TWL model provides a relatively simple tool to assess the increasing exposure of shore-front properties to erosion. However, the actual extent of the erosion depends on the nature of that property, whether it is a foredune, resistant sea cliff, or fronted by a shore protection structure. Simple "geometric" (similar to the Bruun Rule) and/or "process-based" models are available to evaluate the potential retreat of foredunes, which depends directly on the TWL (Komar et al., 1999). Models to predict the extent of sea-cliff erosion during a storm are more complex in that they depend on the resistance of the cliff to the impact forces of the waves (e.g., Collins and Sitar, 2008).

Case Study of Neskowin, OR

Erosion has been particularly acute along portions of the Neskowin littoral cell in southern Tillamook County (see Figure 6.2 for location). This coastal change has been caused in part due to the effects of recent major El Niños as well as the progressive rise in sea level and wave heights over recent decades. Examination of lidar and beach monitoring data for the Neskowin cell documents coastal change between 1997 and 2008 (Figure 6.13). While the effects of the 1997–1998 El Niño winter storms were relatively moderate at high elevations on the beach (dune toe, Figure 6.13), erosion from the 1998–1999 winter was particularly acute adjacent to the village of Neskowin where the beach eroded by as much as 50 m. Similar to the Rockaway cell, the beach monitoring indicates that the southern end of the cell has continued to erode in subsequent years, with some storms having produced as much as 25 m of dune retreat during a single event (Allan and Hart, 2008).

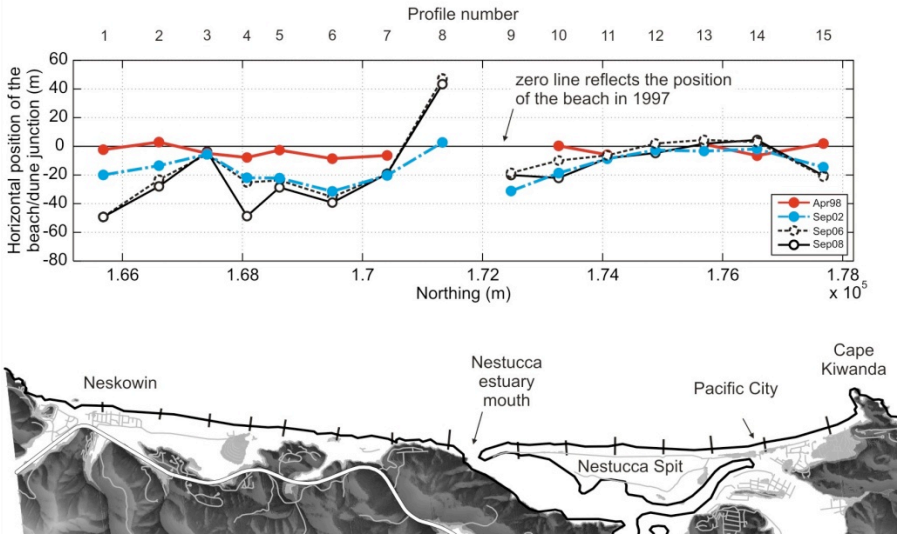


Figure 6.13 Positional changes in the beach/dune toe (elevation of 6 m) along the Neskowin Cell between 1997 and 2008 derived from lidar data and real-time kinematic differential Global Positioning System measured beach profile response. Circles and numbers correspond to the transect locations identified on the map (Allan et al., 2009).

Erosion and flooding along the beaches of Neskowin has today become a harbinger of the probable impacts of future climate change for other coastal communities along the Oregon coast. For example, the loss of significant fronting beach in this area that otherwise previously protected properties built along the shore has substantially increased the potential for catastrophic losses during a major storm. The best example of this scenario was a major storm that occurred on January 5th 2008 that came close to removing oceanfront homes in Neskowin. Because beach elevations had progressively been lowered over the course of the 2007/08 winter the combination of extreme waves during the January storm along with high ocean water levels enabled waves to break close to shore, scouring down the beach face and eventually undermining the toe of a riprap structure and causing part of the structure to fail (Figure 6.14). As a result, given that many beaches in Tillamook County have continued to see very little recovery in the intervening years (i.e., beaches today are narrower and have less sand volume compared with beaches in the mid 1990s), the community of Neskowin in particular remains at

high risk of being impacted by both coastal erosion and ocean flooding in the ensuing winter seasons and from future climate change.



Figure 6.14. Loss of beach at Neskowin has increased the potential for waves to damage coastal structures built to protect properties as well as increased the potential for flooding of homes built behind the structures.

The Neskowin Coastal Hazards Committee (NCHC) is a local community group that has emerged in response to increased coastal hazards along this stretch of the Oregon Coast. The group is chaired by a county commissioner and its mission is to recommend to state and county agencies and officials ways to maintain the fronting beach and protect the community through both short- and long-term strategies and to explore ways to plan for and adapt to the potential future climate induced changes in the Neskowin coastal area. Several state agencies and university researchers are collaborating with the NCHC to find solutions to Neskowin's erosion and flood hazard issues, and identify promising practices and policies that may be relevant to other coastal Oregon communities. The Department of Geology and Mineral Industries (DOGAMI) is responsible for hazards assessment and mapping and the department has been monitoring the Neskowin shoreline for years. Oregon State Parks (OSP) has jurisdiction and permitting authority for shore protection structures and is interested in communities being proactive about planning rather than having to deal with emergency permits during a crisis. The Department of Land Conservation and Development's (DLCD) Coastal Management Program supports local planning for hazards (Goal 7), coastal shorelines (Goal 17), and beaches and dunes (Goal 18). Neskowin's leadership in trying to address coastal erosion has been rewarded by a planning grant from DLCD to Tillamook County to conduct a "coastal hazards adaptation plan" for Neskowin and the county. Researchers from Oregon State University, the US Geological Survey (USGS), and Oregon Sea Grant are working on a separate but related project to evaluate the climate impacts of shoreline erosion, and assess the vulnerability of communities to those hazards. Findings from this work are informing the Neskowin Coastal Hazards Committee and the Tillamook County Coastal Hazards Adaptation Plan.

Early successes of the NCHC include the official formation of the group and recognition by Tillamook County, monthly meetings with formal agendas and minutes (Figure 6.15), drafting and sending a letter to community residents describing the nature of the hazards and why the group has formed, a formal assessment and map by Oregon State Parks evaluating the condition of the riprap along the entire community by individual tax lot, and the collection of research materials on beach protection strategies used by other states. Continued communication and interaction between local groups like this one and various state and federal agencies will be necessary for Oregon's coastal communities to plan for and adapt to coastal impacts due to climate change.



Figure 6.15 Officials from DOGAMI and Oregon Sea Grant presenting scientific information about local coastal hazards to the NCHC in early 2010.

6.5 Climate Change Impacts on Estuaries

The Oregon shoreline is interrupted by 43 estuaries (major and minor) and tidal creeks (estuaries defined as coastal water bodies that had a National Wetland inventory estuarine polygon; Lee and Brown, 2009) that encompass a broad range of land-margin habitats located at the nexus of the land and the sea. The types of estuaries along the Oregon coast are diverse and include: (a) river-dominated drowned river mouths (i.e., Columbia, Umpqua, Coquille, Rogue, Chetco); (b) tidal dominated drowned river-mouths (i.e., Tillamook, Yaquina, Coos); (c) bar-built coastal lagoons (i.e., Netarts, Sand Lake); and (d) numerous tidal creeks (Rumrill, 2006; Lee and Brown, 2009). Physical and biochemical conditions are highly dynamic within the estuaries, and they exhibit considerable spatial and temporal variability that is generally reflective of the extent to which the tidal basins are driven by inputs from coastal watersheds versus nearshore oceanic waters (De Angelis and Gordon, 1985; Sigleo et al., 2005; Sigleo and Frick, 2007; Brown and Ozretich, 2009).

It is anticipated that warming of Oregon's temperate climate will contribute to fundamental changes along the coast including shifts in the timing and intensity of coastal storms (Section 6.3.1), changes in precipitation and the delivery of freshwater inputs (Chapter 1), sea-level rise (Section 6.2), and increased inundation of the shallow tidal basins. Regional coastal climate change may also result in changes in the intensity and timing of coastal upwelling, shifts in temperatures and dissolved oxygen concentrations (Chapter 1), and alteration of the carbonate chemistry of nearshore waters (Chapter 1). The combination of these meteorological and nearshore ocean changes will exert stress on the communities of estuarine organisms and alter water quality conditions in estuaries. The range of estuarine community responses to the climate change stressors may include changes in the abundance, distribution, growth, and reproduction of submerged aquatic vegetation (Short and Neckles, 1999; Kairis and Rybczyk, 2010; Shafer et al., 2008), disruption of shell formation for calcifying organisms (Miller et al., 2009), alteration of the phenology of phytoplankton blooms (Nixon et al., 2009), shoreward migration or loss of tidal marshes (FitzGerald, et al., 2008), and increased colonization by non-indigenous aquatic species (Stachowicz et al., 2002). It is important to note, however, that these changes are based on data from other regions and there is considerable uncertainty in these projections. In addition, long-term time-series data are lacking for Oregon systems to definitively identify perturbations of the estuarine communities and water quality that can be attributed to anthropogenic climate change.

6.5.1 Inundation of Estuarine Wetlands

The estimated long-term rate of coastal wetland loss from all anthropogenic impacts is greater for the Pacific Coast than for any other area of the U.S. (US EPA, 2008) and an estimated 70-90% of tidal wetland habitat has been altered or lost in Oregon in the past 150 years (Brophy, 2005). Added to these historical impacts, sea level rise (SLR) will result in additional wetland loss with a concomitant loss of associated ecosystem services, such as habitat for juvenile salmon and waterfowl, flood control, maintenance of estuarine water quality, and carbon sequestration (Zedler and Kercher, 2005). While considerable research is needed to determine the precise relationships of these ecosystem services to specific wetland types, the first step is to project how SLR will affect the area and distribution of wetland types in Oregon. Susceptibility of these wetlands to SLR will vary among estuaries due to difference in RSL rise rate along the coast (Figure 6.5) as well as due to differences in the extent and type of wetlands within in each estuary (Lee and Brown, 2009).

Deciding how to begin modeling the vulnerability of estuarine wetlands to SLR in the Pacific Northwest is not clearly presented in the existing literature. However, in a recent review of SLR impact models, Mcleod et al. (2010) recommended the Sea Level Affecting Marshes Model (SLAMM) as an inexpensive yet detailed assessment of vulnerability of wetland habitats to sea-level rise. SLAMM models five of the primary processes involved in wetland conversion and shoreline modification during SLR including accretion, saturation, inundation, overwash, and erosion
(Park et al. 1989; www.warrenpinnacle.com/prof/SLAMM). SLAMM was initially developed for modeling intertidal marshes on the U.S. East coast, but since it is based on National Wetland Inventory (NWI; http://www.fws.gov/wetlands/) it offers a generalized tool for predicting SLR-induced change in freshwater, brackish, and estuarine tidal marsh ecosystems. For

example, SLAMM was applied at a regional scale to evaluate effects of SLR on 11 sites in Puget Sound and along the Pacific Coast from Washington to Tillamook Bay (Glick et al., 2007).

Among its advantages, SLAMM is scalable, GIS-based, and is capable of including the effects of surrounding land cover such as impervious surfaces and coastal armoring (e.g. dikes and seawalls). Additionally, SLAMM is an open-source program, and offers users flexibility in terms of model complexity and climate change scenario choices. Because of its generality and flexibility as a planning decision support tool, SLAMM has gained traction among coastal managers and has been used, for example, to evaluate SLR impacts on ecosystem services of Georgia marshes (Craft et al., 2009a). In addition, several Federal, State and non-government agencies are utilizing SLAMM to model potential effects on tidal wetlands in Pacific Northwest estuaries and aid decisions about their restoration. Model results are being used by the U.S. Fish and Wildlife Service in developing comprehensive conservation plans for many National Wildlife Refuges (NWR) including the Willapa Bay NWR, in Washington and, in Oregon, the Bandon Marsh (NWR), Nestucca Bay NWR, and Siletz Bay NWR in Oregon. The USGS is using SLAMM to model habitat responses to SLR in the Yaquina Bay Estuary based on 1m, 1.5m, and 2m SLR scenarios. A broader, watershed level SLAMM modeling effort for Oregon estuaries is being undertaken by Ducks Unlimited.

Available SLAMM predictions for Oregon wetland refuges indicate different types of impacts across different estuaries or estuarine segments. Current results indicate that Bandon Marsh NWR is predicted to lose between 19% and 92% of its swamp by 2100 depending on the SLR scenario utilized (Clough and Larson, 2010a). SLAMM simulations for Siletz Bay NWR indicate dry land losses will range from 12%-40% by 2100 depending on the SLR scenario (Clough and Larson, 2010b). In the Nestucca Bay NWR, SLAMM predicts that the non-diked portions are vulnerable to SLR and 7%-30% of the dry land is predicted to be lost (Clough and Larson, 2010c). Preliminary SLAMM results for the Yaquina Estuary based on 30 m digital elevation models (DEMs), recently updated NWI data, and a 1 meter SLR scenario by the year 2100, with no protection to developed areas, indicates a 74% reduction of tidal flat area, 94% reduction of irregularly flooded marsh, and a 85% increase in regularly flooded marsh from their initial areas (Reusser et al., unpublished results).

It is important to recognize the limitations of SLAMM, both because of general model limitations and because of data gaps for Oregon estuaries. There are limited published peer reviewed studies including the SLAMM methodology. Craft et al. (2009a) and the subsequent responses by Kirwan and Guntenspergen (2009) and Craft et al. (2009b) provide valuable insights into gauging the confidence of model simulations due to the uncertainty of input parameters, primarily base elevation and accretion rates. The use of lidar data in SLAMM has been shown to increase certainty of the base elevation parameter, however, precise elevation data collected with lidar are sparse for most of the upland area of the coastal watersheds in Oregon.

Another important data gap is the paucity of accretion rates specific to regularly flooded marsh, irregularly flooded marsh, and the tidal flats. For the Oregon modeling efforts mentioned above, the regional low marsh accretion rates measured in the Salmon River (Thom, 1992) were the best available data. Short-term (1-2 year) accretion rates are currently being measured in 7 Oregon estuaries in both low and high marsh habitat by the U.S. Environmental Protection Agency

(EPA), and will be available in the future for refinement of SLAMM simulations. Another data gap with regard to inundation of estuarine habitats is how marsh accretion rates will change if there are dramatic changes in rates of RSL rise. Some studies from other regions of the U.S. have suggested that marsh habitats can increase their accretion rates to keep pace with sea level rise if there is an adequate sediment supply; however, when the rate of RSL rise exceeds a certain threshold marsh communities will no longer be able to keep up with SLR and will be inundated (Mudd et al., 2009).

One general model limitation is in how salinity changes are modeled, in particular with the assumption of a salt-wedge estuary, which is likely to underestimate the extent of increased salinity experienced by emergent marshes in response to increased advection of sea water into estuaries. Another general model limitation is that SLAMM does not predict changes to submerged aquatic vegetation (SAV), an important habitat type for juvenile salmon and migratory waterfowl.

As with any model, researchers and managers should weigh the uncertainties in SLAMM predictions against their specific needs. We suggest that the current version of SLAMM is sufficient to address general questions about changes in emergent wetland in the Pacific Northwest, but not sufficient for higher resolution projections due both to regional data gaps and model assumptions. Through a joint research project, the U.S. EPA and USGS are working toward filling these gaps with new data collection for accretion rates, and are developing estuarine salinity models, and software enhancements to the SLAMM program. These regionalized input data and model improvements should allow us to more accurately model changes in critical wetland habitats in Oregon estuaries in the next few years.

6.5.2 Status of Climate Change Research on Estuarine Water Quality

There have been relatively few studies on water quality conditions in Oregon estuaries, with most of the research focused on the Columbia, Coos, Yaquina, and Tillamook estuaries. There are even fewer long-term estuarine datasets to assess the effect of climate change on Oregon estuaries, with the exception of the Columbia River Estuary. High variability in water quality conditions combined with differences in sampling in historic datasets may make it difficult to assess climate change impacts in estuaries even when long-term datasets exist. Little or no research has been conducted on the effect of climate change in most Oregon estuaries, with the Columbia River Estuary being an exception.

Even when long-term datasets are available, it is often difficult to distinguish climate impacts from other anthropogenic modifications. Jay and Naik (2002) estimated that about 9% of the flow reduction in the Columbia River is related to climate change and 7% is associated with water withdrawals. Historical changes in estuarine currents, salinity and sediment transport remain poorly understood, and a combination of approaches including historical data analyses, numerical modeling, and new observations have been recommended to elucidate the effect of climate change on estuaries (Jay and Naik, 2002). Research is currently being conducted on the effect of climate change on the Columbia River Estuary by the Center for Coastal Margin Observation and Prediction (http://www.stccmop.org/). Numerical modeling is being

conducted, which simulates the effect of changes in SLR and river flow using climate scenarios provided by the Climate Impacts Group at the University of Washington. Response variables that are being used to assess the effect of climate change include estuarine salinity intrusion, plume volume, and plume area. Model simulations have found that the effects of climate change on the Columbia Estuary are spatially variable and that salinity intrusion length in the estuary is sensitive to climate drivers (personal communication with Dr. Antonio Baptista). In addition, model simulations suggest that there is high degree of non-linearity in estuarine responses to climate drivers. Since there has been little research on the effect of climate change for most Oregon estuaries, in this section we will review the key climate drivers influencing water quality parameters, and will speculate as to how climate change may influence estuarine water quality.

6.5.3 Factors that Influence Water Quality in Estuaries

Water quality conditions in PNW estuaries are strongly influenced by both freshwater inflow and ocean conditions (Roegner and Shanks, 2001; Brown and Ozretich, 2009; Lee and Brown, 2009; Roegner, et al., 2010). As a result, there is the potential for climate change to influence estuarine water quality. During the wet season (November to April), water quality conditions in estuaries are dominated by freshwater inflows, while during the dry season (May to October), freshwater inflows to the estuaries decline and the estuaries become more marine-dominated. The dry season also roughly coincides with the upwelling season on the Oregon shelf (Chapter 1). Previous studies have demonstrated that water quality conditions within PNW estuaries during the summer are influenced by intrusions of oceanic water into the estuaries, affecting nutrients (Haertel et al., 1969; de Angelis and Gordon, 1985; Brown and Ozretich, 2009; Lee and Brown, 2009), phytoplankton (Roegner and Shanks, 2001; Roegner et al., 2002; Brown and Ozretich, 2009; Lee and Brown, 2009; Roegner et al., 2010), and dissolved oxygen levels (Pearson and Holt, 1960; Haertel et al., 1969; Brown and Power, in review). Water quality conditions in Oregon estuaries are also influenced by land use/cover and human activities in the watersheds. Climate change has the potential to influence water quality conditions in estuaries primarily through changes in precipitation in the watersheds, land cover in the watersheds, temperature (atmospheric, riverine, and oceanic), coastal upwelling, wind stress, and RSL rise. Changes in snow melt will primarily affect the Columbia River Estuary (Hamlet and Lettenmaier, 1999), because of the small amount of snow pack in the watersheds of the coast range.

Different types of estuaries will have differing responses to climate change drivers (Table 6.1). For example, a lagoonal estuary with limited freshwater inflows, such as Netarts estuary, will probably have relatively small changes in salinity; however, this system will be strongly influenced by conditions occurring on the inner shelf and any changes in upwelling related to climate change will likely influence water temperature, nutrients, dissolved oxygen, and pH levels in this type of estuary. Netarts Estuary may also be susceptible to coastal erosion on the barrier spit, which forms the estuary. In comparison a river-dominated estuary, such as the Columbia River Estuary, may experience changes in the salinity regime in response to SLR and changes in freshwater runoff and snow melt.

There are some indications that there has been an increase in upwelling favorable wind stress during the last 50 years (Chapter 1); However, global models for different emission scenarios predict there will little change in upwelling favorable wind stress in the PNW (Chapter 1). Since

coastal upwelling is a major driver in water quality in Oregon estuaries, and there are unknowns with regard to future changes in coastal upwelling, in the following sections, we include discussions of how water quality conditions may be influenced by changes in upwelling.

The most common causes of water quality impairments in Oregon estuaries are temperature, dissolved oxygen, and bacterial contamination (Oregon Department of Enviromental Quality, Oregon's 2004/2006 Integrated Report Database, http://www.deq.state.or.us/wq/assessment/rpt0406/search.asp). Climate change has the potential to cause water quality impairments through various mechanisms that will be discussed in the following sections. We will review the possible effects of climate change on estuarine water temperature, salinity, dissolved oxygen, nutrients, chlorophyll a, bacterial contamination, and carbonate chemistry and the key climate drivers which influence these water quality variables.

Table 6.1. Hypothesized effects of climate alterations on estuaries.

Climate Alteration	Potential Effects
Sea level rise and extreme water levels.	Increased inundation of estuarine habitats including tidal flats, marshes, and SAV. Increased intrusion of oceanic water into estuaries. Extent of effect will vary with relative river flow, location in the estuary, and RSL rise rates in the vicinity of the estuary.
Increased winter precipitation and decreased summer precipitation	Increased winter–early spring flow of coastal rivers and creeks and reduced flow during summer. Extent of effect related to relative river flow, with a greater impact on river-dominated estuaries (e.g., Umpqua) and tidal coastal creeks (e.g., Yachats) than on tide-dominated estuaries (e.g., Yaquina, Coos), with smallest effect on bar-built estuaries (e.g., Netarts).
Increased snow melt	Change in seasonal pattern of river flow into the Columbia River; minor changes in other coastal estuaries and creeks.
Increased air temperature	Potentially high vulnerability of intertidal organisms because of the high proportion of intertidal area in Oregon estuaries that may be exposed to elevated temperatures. Air temperature also has the potential to influence water temperatures particularly in the upriver portions of estuaries.
Increased upwelling	Increased advection of high nutrient ocean water into the lower estuary during summer. Possible increase in the advection of low dissolved oxygen and low pH water into the lower estuary during the summer. Changes associated with upwelling may be more important in tide- versus river- dominated estuaries.
Increased storm activity	Potential breaching of barrier dunes at mouth of estuaries without jetties (e.g., Alsea, Siletz) and episodic input of sediment to estuaries. Estuaries with jetties may be less impacted (e.g., Yaquina, Coos, and Rogue).
Ocean acidification	Unknown effect on estuaries or how alterations may vary across estuary classes.

6.5.4 How Climate Change May Influence Estuarine Water Quality and Research Needs

6.5.4.1 Temperature

Estuarine water temperature is an important factor influencing the rates of biological, chemical, and physical processes in estuaries as well as the distribution and well being of aquatic life. The effect of global climate change on estuarine water temperature will vary with estuary configuration, estuary type, and location in the estuary. The primary climate drivers which may influence estuarine water temperature are increases in the temperature of riverine inflow, air and

the ocean, SLR, changes in upwelling, and changes in river discharge. The Oregon water temperature criterion for ocean and bay water states that these waters may not be warmed by more the 0.3 ˚C above natural conditions (ambient levels) unless a greater increase would not reasonably be expected to adversely affect fish or other aquatic life (DEQ, 2008). More research is needed to determine if global climate change would result in elevation of estuarine water temperatures above this threshold.

In most Oregon estuaries during the dry season, there is a longitudinal gradient in water temperature (Brown et al., 2007; Roegner et al., 2010). Cooler temperatures typically occur near the estuary mouths and increase to a maximum in the tidal freshwater regions. Figure 6.16 shows the longitudinal water temperature gradient in the Yaquina Estuary. Both historic (1960-1985) and recent data from the Yaquina Estuary (1986-2007) are presented in Figure 6.16. These data do not suggest that there have been major changes in estuarine water temperature. However, formal analyses were not performed due to differences in sampling (e.g., depth and tidal stage) within the compiled datasets. In the upriver portion of the estuaries, instantaneous water temperatures at times exceed temperature criterion for the protection of migration and juvenile rearing of salmon and trout (Figure 6.16). The relatively cool temperatures near the mouth of the estuaries result from the intrusion of cold oceanic water into the estuaries.

Figure 6.17 shows an example of the coupling between inner shelf water temperature and flood-tide water temperature near the mouth of the Yaquina Estuary. Water temperatures in the lower portion of Oregon estuaries are temporally variable during the upwelling season and respond rapidly to changes in alongshore wind stress (Figure 6.17; Brown and Ozretich, 2009). Long-term measurements of ambient water temperature have also been collected by a series of dataloggers deployed within the South Slough estuary (a sub-system of the greater Coos Bay) since 1995 (Rumrill, 2006). Water temperatures inside Coos Bay are positively correlated with the Pacific Decadal Oscillation (PDO; r^2=0.55), and water temperatures inside the estuary provide a general reflection of fluctuating regional cycles in the nearshore Pacific Ocean waters (Figure 6.18; O'Higgins and Rumrill, 2007).

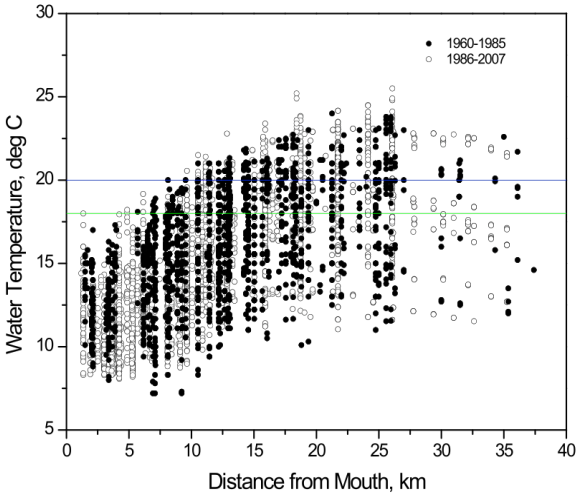


Figure 6.16 Dry season (May - October) water temperature versus distance from the mouth for the Yaquina Estuary (Data source: Brown et al., 2007). The blue line shows the maximum temperature criterion recommended for protection of migrating salmon and trout, and the green line shows the criterion for protection of migrating and non-core juvenile rearing of salmon and trout (US EPA, 2003).

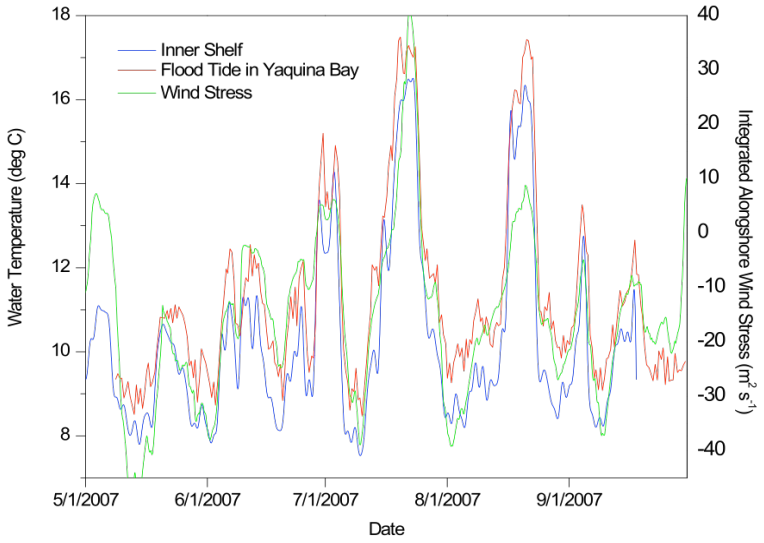


Figure 6.17 Flood-tide water temperature in Yaquina Bay, inner-shelf water temperature, and alongshore wind stress. Water temperature in Yaquina Bay is significantly correlated with inner shelf water temperature ($r = -0.97$, $p<0.001$) and alongshore integrated wind stress, an indicator of upwelling ($r = 0.85$, $p < 0.001$). Yaquina Bay data are from the US EPA and inner shelf data are from Station SR15, which is located 12 km south of Yaquina Bay at the 15-m depth contour (Data Source: http://www.piscoweb.org). Alongshore integrated wind stress (normalized by density of seawater) is calculated following the method of Austin and Barth (2002), using wind data from Station 46050 with a decay coefficient of 6 days.

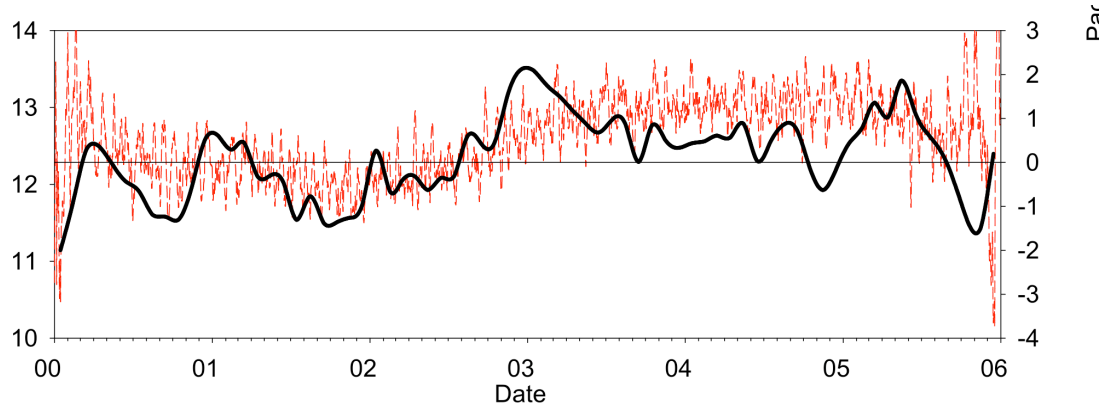


Figure 6.18. Time-series of estuarine water temperature within the South Slough estuary (red) and the Pacific Decadal Oscillation (PDO) index (black) over the period 2000-2006 (r^2 = 0.55, $p < 0.01$); from O'Higgins and Rumrill, 2007.

Most of the assessments of trends in upwelling in the California Current system have relied upon trends in wind stress or coastal upwelling index (Chapter 1) or model simulations of the effect of different emission scenarios (Chapter 1; Mote and Salathé, 2010; Wang et al., 2010; Mote and Mantua, 2002; Snyder et al., 2003). Due to the high variability in water temperature in the coastal ocean, and the lack of long-term high resolution datasets, it is difficult to assess whether there have been changes in nearshore coastal upwelling (Chapter 1). However, there are some relatively long-term estuarine water temperature datasets available. The close correspondence between estuarine flood-tide water temperatures and inner shelf water temperatures demonstrates that these estuarine datasets may be useful to assess long-term trends in upwelling.

Figure 6.19 shows average dry season flood-tide water temperature near the mouth of the Yaquina Estuary. The interannual patterns in average dry season flood-tide water temperature (using data from 1992-2010) are significantly correlated with cumulative alongshore wind stress ($r = 0.58$, $p < 0.01$; source for cumulative alongshore wind stress is http://damp.coas.oregonstate.edu/windstress/), and are an indicator of the amount of upwelling occurring during May through September on the inner shelf. During 1992-2010, there is a significant decreasing trend in dry season flood-tide water temperature at the Yaquina Estuary. Since cool water temperatures indicate upwelling activity, these data suggest that there has been a long-term increase in coastal upwelling during the last 19 years; however, it isn't clear if this is due to climate change or long-term basin-scale oscillations. More research is needed to compile additional data that may be available, and to explore relationships with climate indices.

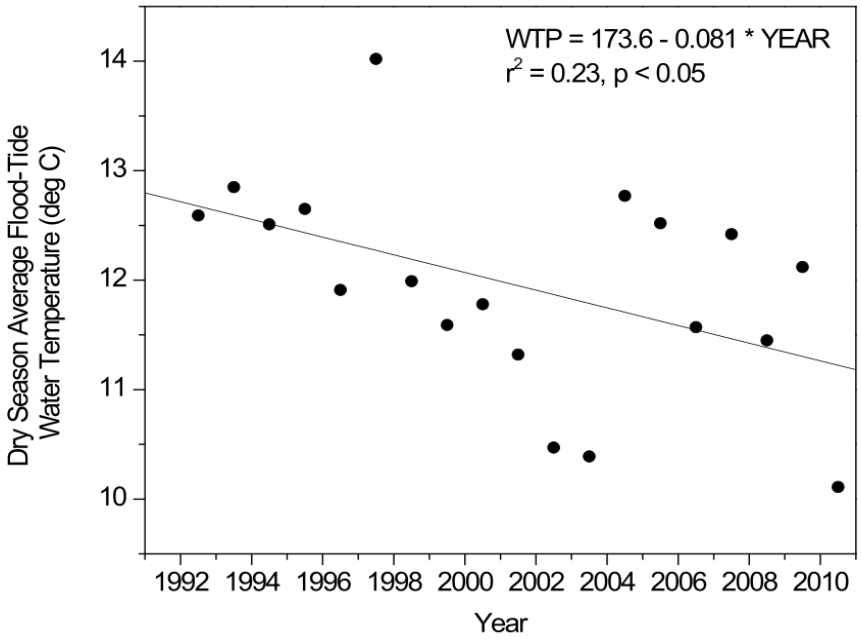


Figure 6.19 Dry season average flood-tide water temperature at the Yaquina Estuary as an indicator of coastal upwelling. Data Source: US EPA and NOAA South Beach tide gauge. Data compiled by Cheryl Brown, US EPA.

In the marine-dominated portion of Oregon estuaries, it may be difficult to separate variability associated with upwelling and downwelling dynamics and longer term cycles (such as Pacific Decadal Oscillation) from a climate change signal. Any increases in upwelling associated with climate change may obscure warming trends in the marine-dominated portion of the estuary at least during the summer. Water temperature in estuaries will be more sensitive to changes in upwelling, than to changes in ocean sea surface temperatures (SSTs) such as those projected by global climate change models (Chapter 1). As discussed in Chapter 1, the climate model projections for SSTs represent surface waters about 100 km offshore, rather than the nearshore which is influenced by upwelling of water from depths that are not as immediately influenced by climate change. An increase in sea level may result in increased intrusion of cool ocean water into the estuaries, further obscuring the climate change signal.

Climate change may result in an increase in water temperatures in the brackish region of the estuaries due to increases in water temperature of freshwater inputs to the estuary, increases in air temperature, and reductions in freshwater inflow during the summer. More research is needed to determine to what extent climate change may influence estuarine water temperatures. Since estuarine water temperatures are closely coupled to upwelling dynamics, improvement is needed in models which are used to predict changes in upwelling (Chapter 1). Since present water temperatures in the upriver portions of estuaries often exceed temperature criterion for protection of salmon and trout, there is the potential for climate change to result in an increase in temperature impairments, and thus to potentially affect the distribution and survival of cold water species.

6.5.4.2 Salinity

Salinity is an important factor influencing the distributions of organisms in estuaries as well as important factor determining the water density in estuaries. Studies in other regions have demonstrated that climate change has the potential to influence estuarine salinity (Gibson and Najjar, 2000; Hilton et al., 2008). Estuarine salinity may be influenced by climate change primarily through changes in precipitation and snow melt in the watershed (resulting in changes in freshwater inflows) and increased intrusion of seawater associated with rising sea levels. In Oregon estuaries, salinity levels increase during the dry season due to reduced freshwater inflows. Rising sea level is expected to have more of an impact on estuarine salinities during the dry season, than during the wet season. The effect of climate change on estuarine salinity will vary with location inside the estuary and the magnitude of the RSL rise rate in the vicinity of the estuary (see Figures 6.3 and 6.5). For example, salinity intrusion due to sea level rise may be more important for Yaquina and Coos estuaries than for the Columbia River estuary, due to differences in land movements. In addition, whether an estuary is river- or tide-dominated may influence the extent of variations in salinity associated with climate change.

As discussed previously in Section 6.5.2, the Center for Coastal Margin Observation and Prediction is conducting modeling on the effect of climate change on salinity in the Columbia River Estuary. Model simulations are showing that salinity intrusion in the Columbia River Estuary is sensitive to climate drivers. The US EPA is presently conducting research on the effect of sea level rise on salinities in the Yaquina, Netarts, Coquille, and Coos estuaries. Figure 6.20 shows simulation results of the effect of sea level rise at three locations in the Yaquina Estuary for different steady river discharges. These simulations show the non-linearity in salinity response to rising sea level. Near the mouth of the estuary (Figure 6.20a), sea level rises of 30–60 cm will result in changes in salinity of less than 2 psu. At this location, there will be little change in peak salinities with rising sea level, while median and 25th percentile salinities will increase with increasing river discharge. At the mid-estuary location (Figure 6.20b), an elevation

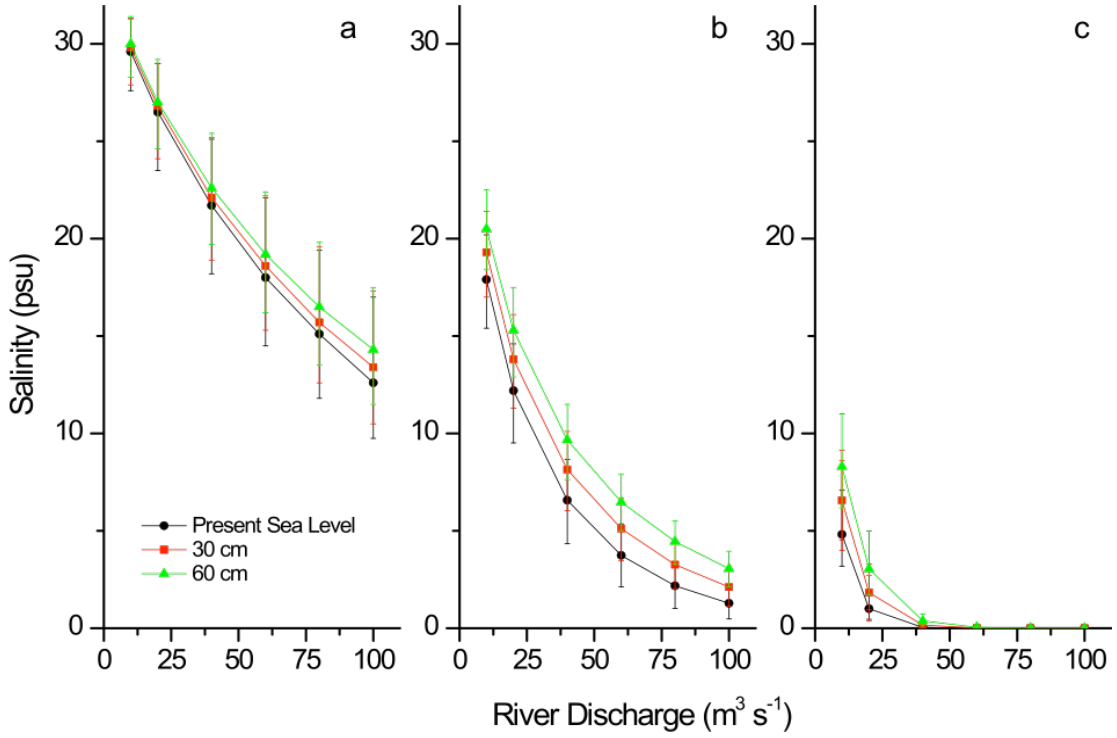


Figure 6.20 Simulated effect of sea level rise on salinity in the Yaquina Estuary for different steady river discharge conditions at three locations (a-c). Simulations were performed for present sea level and sea level rise of 30 and 60 cm. The symbol indicates the median simulated salinity and the lower and upper error bars show the 25th and 75th percentiles. Panels a-c show model results for locations that are 10, 19, and 26 kms, respectively, from the mouth of the estuary. Source: Cheryl Brown (US EPA).

of sea level of 60 cm will increase salinities by about 2 -3 psu across a broad range of river discharge levels. In the oligohaline portion of the estuary (Figure 6.20c), salinity increases associated with a 60 cm rise in sea level are only present at low discharge rates. These simulations show how the interaction of changes in river discharge and sea level associated with climate change produce non-linear patterns in the response of estuarine salinity, which vary with location inside the estuary. Future work by the US EPA will link estuarine models to down-scaled climate scenarios to examine the effect of RSL rise on estuarine salinity for systems ranging from lagoonal system (e.g., Netarts) to a river-dominated (Coquille) system.

6.5.4.3 Dissolved oxygen

Dissolved oxygen levels within estuaries result from the interaction of numerous physical and biological factors, including stratification and mixing, water column and benthic oxygen demands, and photosynthesis. Dissolved oxygen levels in estuaries have the potential to be influenced by climate change through numerous climate drivers including temperature (of river, air, and ocean), riverflow, and ocean conditions.

Dissolved oxygen (DO) is an important water quality metric because of its effects on the well-being of resident and transitory estuarine organisms. Hypoxia is commonly defined as occurring when the DO levels fall below 2 mg l^{-1}; however, biological stress for organisms has been documented at DO levels between 2 and 5 mg l^{-1} (U.S. EPA, 2000; Diaz and Rosenberg, 1995).

The existing State of Oregon dissolved oxygen criterion for estuaries (6.5 mg l^{-1}) is based on a review of physiological requirements of biota, and is high compared to DO criteria for other estuaries (U.S. EPA, 2003). A review of the DO criterion found that 6.5 mg l^{-1} may be difficult to achieve in Oregon estuaries during the summer due to natural conditions (DEQ, 1995).

In Oregon estuaries, typically low dissolved oxygen conditions have been observed near the upper end of salt water intrusion (DEQ, 1995), such as in the sloughs of Tillamook Estuary. In the Yaquina Estuary, there has been a shift in the location of low dissolved oxygen conditions. Historically, low dissolved oxygen levels occurred in the upper portion of the Yaquina Estuary (20 km from the mouth; Brown et al., 2007), which coincided with the location of the point source discharges and a region of extensive log rafting. Recently, there have been occurrences of severe hypoxia on the inner continental shelf of Oregon (Chapter 1; Grantham et al., 2004; Chan et al., 2008), which influence dissolved oxygen levels in Oregon estuaries. For example, low dissolved oxygen conditions have been periodically imported from the inner shelf into the Yaquina Estuary (Brown et al., 2007), as well as into Umpqua, Tillamook, Coos, and Siletz estuaries (Brown and Power, in review). This low dissolved oxygen water has a distinctive thermal signature and dissolved oxygen levels in the Yaquina Estuary are correlated with alongshore wind stress. Recent time-series data (2001-2007) demonstrate that the water advected into the Yaquina Estuary from the coastal ocean has dissolved oxygen levels <5 and <6.5 mg l^{-1} about 13% and 38% of the time, respectively (Brown and Power, in review). Research has also demonstrated that dissolved oxygen levels in South Slough Estuary are related to ocean conditions. O'Higgins and Rumrill (2007) found a negative correlation between dissolved oxygen levels inside the South Slough and the PDO (O'Higgins and Rumrill, 2007).

Any changes in frequency or intensity of upwelling may cause changes in the occurrence of low oxygen conditions in the marine-dominated portion of estuaries, particularly for tide-dominated estuaries. A rise in sea level may cause oceanic low dissolved oxygen water to penetrate further into the estuaries. In the more upriver parts of the estuaries, dissolved oxygen levels may also be influenced by climate change through increased stratification, reduced flushing, as well as temperature effects. In estuaries, dissolved oxygen levels typically decrease with increasing temperature due to both reduced solubility of oxygen in water and to increased respiration and decomposition rates. In a comparison of water quality conditions in four west coast estuaries (including Coos Estuary), the number of hours of oxygen stress (dissolved oxygen levels < 5 mg l^{-1}) per day increased linearly with mean annual water temperature (O'Higgins and Rumrill, 2007). Through various mechanisms, climate change has the potential to influence dissolved oxygen levels in estuaries; although to what extent this may happen is unknown at this time.

6.5.4.4 Nutrients

Oregon estuaries receive nutrients from watershed point and non-point sources as well as from the coastal ocean (Haertel et al., 1969; de Angelis and Gordon, 1985; Sigleo et al., 2005; Sigleo and Frick, 2007; O'Higgins and Rumrill, 2007; Brown and Ozretich, 2009; Lee and Brown, 2009). There is considerable interannual variability in coastal upwelling and in nutrient concentrations on the shelf (Corwith and Wheeler, 2002; Wheeler et al., 2003), as well as in the input of oceanic nutrients to coastal estuaries (Brown and Ozretich, 2009). In addition, previous studies have suggested that the presence of red alder trees in the watersheds is a significant source of nitrogen

to many Oregon estuaries (Wigington et al., 1998; Compton et al., 2003; Naymik et al., 2005; Brown and Ozretich, 2009; Lee and Brown, 2009).

Climate change has the potential to influence nutrient levels in Oregon estuaries through various mechanisms. If there are changes in upwelling, the nutrient input (primarily as nitrate and phosphate) from the coastal ocean may change. Sea level rise may cause oceanic nutrients to propagate further into the estuary. Previous studies have shown that dissolved inorganic nitrogen loading to Oregon estuaries is related to river discharge (Sigleo and Frick, 2007; Brown and Ozretich, 2009); therefore, changes in precipitation and resulting riverflow may influence nutrient loading. Nutrient utilization inside estuaries will be influenced by changes in estuarine stratification, flushing and residence times, and the timing and magnitude of nutrient delivery. Previous studies have suggested that there is little utilization of nutrients inside Oregon estuaries during the wet season due to high flushing rates and low solar irradiance (Haertel et al., 1969; Colbert and McManus, 2003; Brown et al., 2007; O'Higgins & Rumrill, 2007). Therefore, estuaries may be more susceptible to changes in nutrient loading and physical effects during the dry season. More research is needed on the effect of climate change on nutrient loading from both oceanic and watershed sources and nutrient utilization inside Oregon estuaries. Research is also needed on how the distribution of red alder may change in response to climate drivers in the watersheds of Oregon estuaries.

6.5.4.5 Chlorophyll a

Previous studies have demonstrated that chlorophyll a is advected into estuaries along the Oregon and Washington coasts from the coastal ocean during the dry season (Roegner and Shanks, 2001; Roegner et al., 2002; Brown and Ozretich, 2009; Roegner et al., 2010). Although the import of chlorophyll a from the coastal ocean to estuaries is related to coastal upwelling, there is a lag between upwelling on the coast and import of phytoplankton to estuaries (Brown and Ozretich, 2009). Ohana-Richardson (2007) further investigated statistical relationships between cell concentrations of the diatom *Pseudo-nitzschia* in South Slough estuary and estuarine water temperatures, and he found that the diatom concentrations were elevated during cold water periods (r^2=0.65, $p<0.05$). Conversely, diatom cell concentrations decreased during periods of estuary warming. The advection of phytoplankton from the coastal ocean into estuaries appears to be more important for tide-dominated than river-dominated estuaries (Lee and Brown, 2009).

A rise in sea level may result in chlorophyll a of oceanic origin reaching further into Oregon estuaries, particularly for tide-dominated systems. As discussed in Section 7.4.1, more research is needed on the effect of global climate change on coastal phytoplankton levels, and the subsequent import of these blooms into the estuaries. In the Yaquina Estuary, there are high chlorophyll a levels in both the high salinity and tidal fresh portions of the estuary (Brown et al., 2007). In the Columbia River Estuary, the main source of chlorophyll a in the spring is riverine sources, and in the summer when river discharge decline oceanic sources increase in importance (Roegner et al., 2010). Newton and Horner (2003) demonstrated that high productivity phytoplankton blooms were imported from the coastal ocean and that these blooms consisted of species of oceanic origin, while moderate blooms which included species of oceanic and estuarine origin also occurred within the estuary. Climate change can also influence chlorophyll a levels in estuaries through changes in nutrient loading, timing in nutrient delivery, and utilization inside the estuary. During the wet season, chlorophyll a levels in Oregon estuaries are

low due to high flushing rates (Roegner et al., 2010) and low solar irradiance, while during the dry season chlorophyll a levels increase (Haertel et al., 1969; Karentz and McIntire, 1977; Colbert and McManus, 2003; Brown et al., 2007; O'Higgins & Rumrill, 2007). Chlorophyll a levels in estuaries will be more sensitive to changes in river inflow during the dry season. Reductions in river discharge during the dry season may result in an increase in chlorophyll a levels in the estuary due to increased residence time of nutrients in the estuary.

6.5.4.6 Bacterial contamination

Climate change has the potential to influence bacterial contamination in estuaries. In some Oregon estuaries, such as Tillamook Estuary, agricultural lands are currently protected from inundation by dikes and levees. However, increases in RSL rise rate and storms may result in increased flooding and inundation of agricultural lands, which may cause an increase in runoff from these lands. As an example, in the Tillamook Estuary high concentrations of fecal coliform levels (indicator of fecal contamination) occurred in the fall and were preceded by dry conditions and high intensity rainfall (Sullivan et al., 2005). Climate projections for the PNW suggest that there may be decreases in summer precipitation and increases in winter precipitation (see Section 2.3.3 in Chapter 1). With this projected change in precipitation pattern, it is possible that there may be higher fecal coliform levels during the fall and early winter if there are sources in the watershed.

In addition, the upwelled waters that are brought to the surface along the Oregon coast (cold, hypoxic, nutrient-rich) are highly conducive to outbreaks of the pathogenic bacterium *Vibrio tubiashii* (Elston et al., 2008), which are lethal to oyster larvae and early juveniles. During 1998, 2006, and 2007, there were losses of oyster larvae and juveniles along the Pacific coast of North America, which was attributed to an outbreak of *V. tubiashii* (Elston et al., 2008). It is believed that these vibrio outbreaks were related to seeding of vibrios and nutrients in the nearshore region associated with coastal upwelling, combined with subsequent elevated water temperatures, which resulted in a rapid increase in vibrio populations (Elston et al., 2008). Thus, changes in upwelling associated with climate change combined with warmer nearshore and estuarine water temperatures may result in an increase in outbreaks of *V. tubiashii.*

6.5.4.7 Carbonate Chemistry

Long-term observational records and empirical studies provide strong evidence that global increases in atmospheric carbon dioxide (CO_2) concentrations have contributed to widespread changes in the carbonate chemistry of ocean waters (Feely et al., 2004; Fabry et al., 2008; Dore et al., 2009). Elevated atmospheric CO_2 values are directly coupled with increased pCO_2 values in the ocean, and the shift in equilibrium results in increased carbonic acid (H_2CO_3) and decreased pH values. Long-term decreases in pH values (ocean acidification) have the potential to pose a major environmental problem, and ocean acidification has emerged as a pressing regional, national, and international issue that has important physiological and ecological implications for marine organisms throughout the world. The biochemical process that contributes to ocean acidification is driven by availability of hydrogen ions (H^+), and shifts in pH values are of particular concern for species such as coral reefs, pteropods, coccolithophores, echinoderms, mollusks, and the larval stages of marine invertebrates that have calcium carbonate skeletons (i.e., calcite, aragonite; Orr et al., 2005; Kurihara, 2008). In the eastern Pacific Ocean, it is

estimated that pH values have decreased by 0.08 units over the past 200 years (Orr, et al., 2005) and will continue to decrease at an annual rate of -0.0019 + 0.0002 pH units per year (Dore et al., 2009).

Recent oceanographic measurements demonstrate that acidified seawater is currently upwelling close to the Pacific continental shelf of Oregon and northern California (Feely et al., 2008). The acidified waters are presumed to move by advective transport directly into Oregon estuaries. This acidified water may have adverse impacts on the larvae of native Olympia oysters (Ostrea lurida) and non-native Pacific oysters (Crassostrea gigas) that inhabit the intertidal zone of Netarts Bay (Langdon and Hales, pers. comm.). Like the larvae of several other groups of marine invertebrates that incorporate calcite or aragonite into their tissues (Orr et al., 2005; Kurihara, 2008), oyster larvae are sensitive to acidified marine waters which contribute to dissolution of their thin calcified shells (Miller et al., 2009).

The presumption that ocean acidification and intensified upwelling is uniformly problematic for shellfish populations in all Oregon estuaries may not be valid, however, because it is not clear how the pH and total alkalinity of estuarine waters are related or influenced by the carbonate chemistry of nearshore ocean waters. For example, water conditions in Coos Bay and the South Slough oscillate seasonally between being driven by watershed inputs during the wet season and ocean inputs during the dry season. It is likely that pH variability and aragonite saturation state in many of Oregon's estuaries differ substantially from the aragonite state in the nearshore ocean, and that ecological conditions for the sensitive early life-history stages of oysters (i.e., pediveliger larvae and settled post-larvae) and other calcifying organisms are only indirectly coupled to the undersaturated carbonate waters further offshore. The carbonate chemistry of Oregon's estuarine waters will be modified on a site-specific basis depending on the extent of watershed influence and biochemical processes that occur within each individual estuary. Furthermore, it is also likely that pH variability differs even more substantially from the ocean waters within the mesohaline and polyhaline regions of the estuary which are influenced by the watershed drainage basin and are critical for larval settlement, recruitment, and early growth of native oysters and other shellfish.

To address the issue of shifting carbonate chemistry within the Coos Bay/South Slough estuary, time-series measurements were examined for water column parameters recorded by a YSI-6600 EDS multi-parameter datalogger equipped with a YSI 605091 pH/ORP sealed gel probe (resolution 0.01 pH unit; accuracy ± 0.2 pH unit; Figure 6.23). The datalogger was deployed within the primary tidal channel of the South Slough estuary (Charleston) where it operated continuously over the period of 2002-2009. Estuary pH values typically ranged between 7.7 and 8.3 throughout each day within South Slough, and we observed a strong tidal signal as well as a diurnal cycle with lowest pH values in mid-morning and highest pH values in mid-afternoon. The daily pH cycle appears to be driven by photosynthesis and respiration of phytoplankton, macroalgae, and submerged aquatic vegetation within the estuary (Figure 6.21). Preliminary analysis reveals that there has been a long-term shift in estuary pH values from a median value of 7.9 in 2002 to a median value of 8.1 in 2009 (Figure 6.22). The intermediary years (2003-2008) exhibited intermediate pH values and the time-series trend is strongly directional (Figure 6.23).

Although the apparent shift toward increased alkalinity falls within the reported accuracy of the pH probe, the directional trend is supported by over 200,000 data points and suggests that increased alkalinity may be real rather than an artifact. The long-term trend toward increased alkalinity of the marine-dominated South Slough estuary is unexpected, and provides evidence that the relationship between ocean acidification and pH values in the estuary is not straightforward. Unlike Netarts Bay, larval settlement and recruitment was consistently good for Olympia oysters (Ostrea lurida) in Coos Bay over the past 5-6 years, particularly within the mesohaline region of Coos Bay where estuarine water column conditions are only indirectly coupled to upwelling of the nearshore ocean waters (Groth and Rumrill, 2009). Datasets similar to those from Coos Bay are available in the Yaquina Estuary (US EPA). These data show that flood-tide pH during the dry season in the marine-dominated portion of the estuary are closely coupled to the inner shelf and alongshore wind stress.

Very little is known about the inherent variability in pCO_2 concentrations in the nearshore marine waters of the Pacific northwest coast (Wootton et al., 2008), and the relationships between ocean acidification and pH values are poorly understood for the different types of Oregon estuaries. Although substantial data records from water quality monitoring stations, municipal outfall discharge sites, mariculture operations (and other sources) exist to characterize temporal variability and fluctuations in pH values in Oregon estuaries, the datasets are typically generated by different types of instruments, vary significantly in resolution and accuracy, and have not been assembled and synthesized into a coherent characterization of pH dynamics. Despite these limitations, the existing datasets have tremendous spatial and temporal coverage, and may be useful to address the potential implications of regional climate change and ocean acidification in Oregon estuaries. Long-term datasets that span periods of multiple decades are required to fully analyze inherent temporal and spatial variability in Oregon's estuarine water parameters, and to identify possible directional changes in pH values as a prospective response variable that can contribute to increased understanding of climate-induced changes in ocean carbonate chemistry.

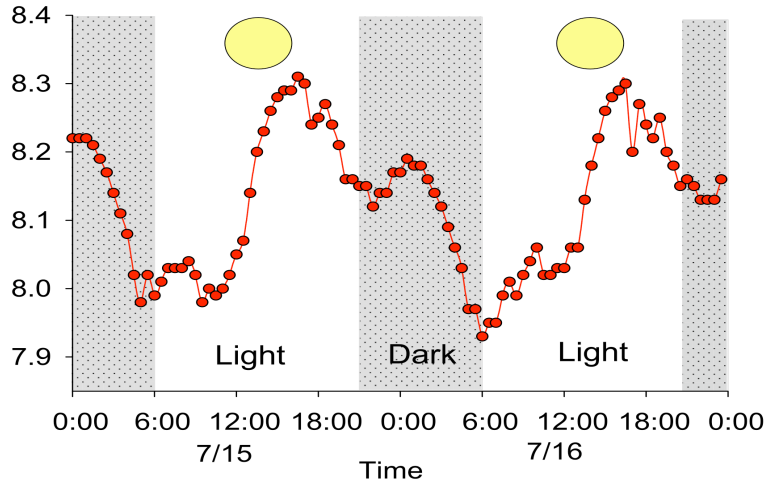


Figure 6.21 Diel cycle of daily changes in pH values measured within the South Slough estuary (Coos Bay, OR) over the period of 15-16 July, 2008, data from South Slough National Estuarine Research Reserve / System-Wide Monitoring Program.

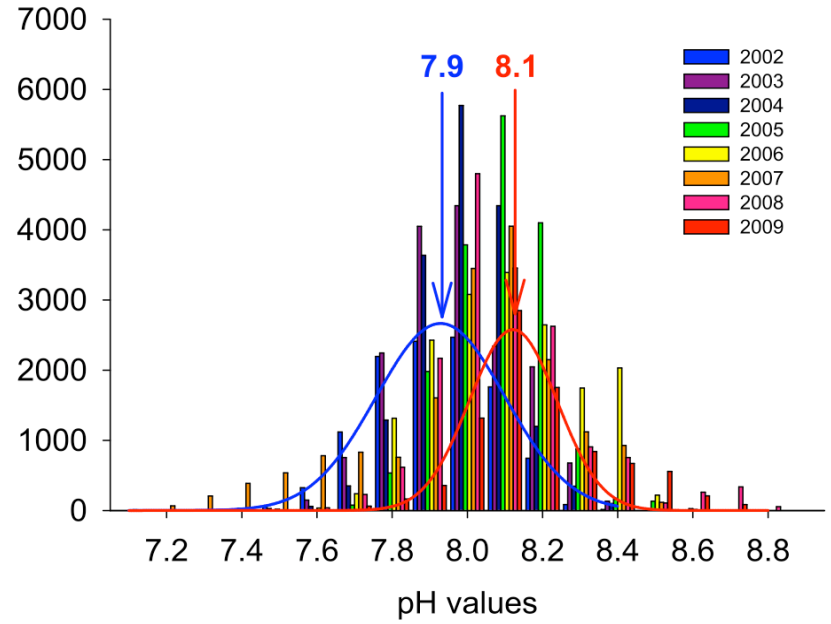


Figure 6.22 Annual frequency distributions for pH values (2002-09) at the Charleston SWMP station, South Slough estuary (Coos Bay, OR). Annual median values are shown for 2002 (blue) and 2009 (red); datasets from South Slough National Estuarine Research Reserve / System-Wide Monitoring Program.

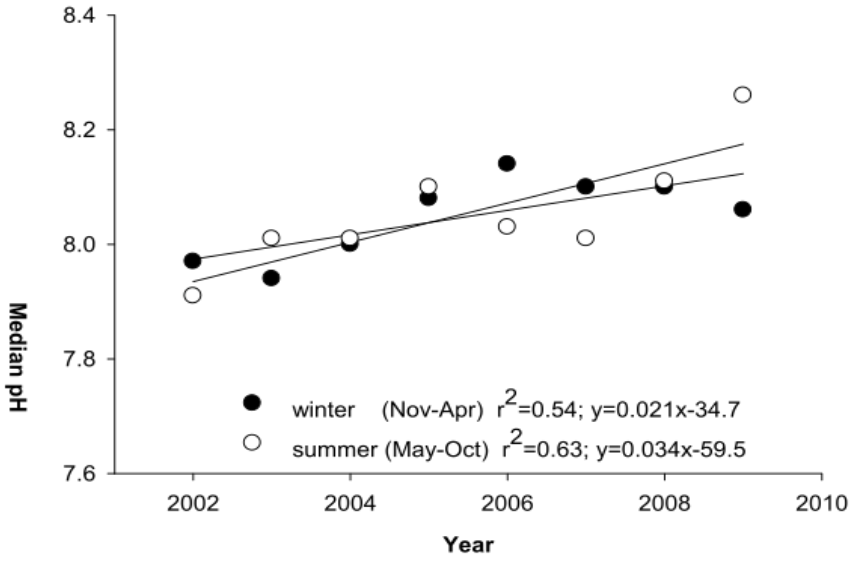


Figure 6.23. Median seasonal pH values at the Charleston SWMP Station (South Slough estuary (Coos Bay, OR) for Winter (Nov-Apr) and Summer (May-Oct). Datasets from South Slough National Estuarine Research Reserve / System-Wide Monitoring Program.

6.6 Climate Change Impacts on Coastal Infrastructure

Climate change may negatively impact Oregon's coastal infrastructure, including both public and private property such as coastal highways, bridges, sewage treatment facilities, airports, oceanfront homes, hotels, industrial sites, and dairy operations around estuaries. As climate conditions change some infrastructure systems may be less effective or may fail altogether, which could alter the function, value, or viability of improvements these systems protect or serve. Public officials and private landowners need to understand how climate change may affect their property and be prepared to make decisions about how to respond and adapt. The sectors described in the following sections may be significantly impacted by climate change (DLCD, 2009).

6.6.1 Transportation and Navigation

Coastal roads, highways, and rail lines are at risk from the effects of increased winter precipitation, increased coastal erosion, and flooding. Over the long term, roads, highways, and railroads will be affected by SLR and increased tidal elevations along the ocean shore, estuaries, and coastal rivers. Airport runways such as those in North Bend and Astoria are located on filled estuarine wetlands and may be at risk of inundation from storm surge and high tides and, over time by SLR. In Oregon there are 11 jettied navigation entrances (including the mouths of the Columbia River, Tillamook Bay, Depoe Bay, Yaquina Bay, Siuslaw River, Umpqua River, Coos Bay, Coquille River Port Orford, Rogue River, and Chetco River) with over 30 miles of rubble mound structures maintained by the Portland District of the US Army Corps of Engineers. Port facilities, jetties and groins will be subject to damage from larger storm waves. Watershed flooding may increase sediment loads into estuaries and thus increase the need for dredging of navigational channels. Increased tidal height will affect docks and bulkheads.

6.6.2 Shore protection and flood control structures

In some coastal counties significant portions of outer coast shorelines have been armored against erosion from ocean waves, primarily in front of properties developed before 1977. Extensive areas in Oregon's estuaries are protected from tidal inundation by dikes, levees, and other structures often to create and protect agricultural lands. Increases in erosion and inundation due to rising sea level and increasing wave heights may threaten the integrity of these coastal structures.

6.6.3 Water supply, wastewater treatment, and stormwater systems

While rainfall in winter is projected to increase, storing water across longer, drier summers may be a problem for some coastal communities where storage systems are already at or over capacity during summer. Reduced precipitation in summer months, especially in conjunction with warmer winter temperatures, may reduce the water available for municipal supply systems. In addition, wastewater treatment facilities are usually located at the lowest elevation in a watershed, which places those facilities at risk from rising sea level and tidal elevation. The capacity of local stormwater management systems may be exceeded as the magnitude or frequency of rainfall events increases, especially as tidal elevations rise leading to localized flooding, accelerated deterioration, and possible system failure. Systems at or near capacity

today may be unable to handle future storm loads, which could have a significant effect on location of future development.

6.6.4 Recreation, travel and hospitality

The Oregon coast has long been an attractive destination for tourists and for those who enjoy its recreational opportunities and resources. Businesses in these sectors may benefit from the effects of climate change on the Oregon coast. Longer, drier summers may benefit Oregon's coastal recreation industries by extending the recreation season and expanding options. However, parks and recreational facilities along the ocean shore and around estuaries are likely to be affected from increased winter storms, ocean flooding, and over time, sea level rise. Coastal forest trails and campgrounds may experience frequent damage from high winds and flooding.

6.6.5 Growth and development

Climate conditions, even though changed from historic patterns, may be relatively attractive compared to other parts of the country and therefore may attract "climate-refugees" seeking more optimum climate conditions. Coastal communities could thus experience increased in-migration, particularly retired persons, which would increase demand for residential and commercial development and demands on public facilities.

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